Annual Report

on

Biomedical Waste Management

for the year 2022

as per

Biomedical Waste Management Rules, 2016

Central Pollution Control Board

(Ministry of Environment Forest & Climate Change)
Parivesh Bhawan, East Arjun Nagar Delhi – 110032

Table of Contents

1. Introduction 4

1.1. Biomedical Waste Management Rules, 2016 4

1.2. Annual Report on Biomedical Waste Management 4

1.3. Status of Annual Report 2022 and Gaps identified 5

1.4. Brief Summary of Bio-medical Waste Management in India 5

1.5. Comparison of BMWM in 2021 and 2022 6

2. Status of Healthcare Facility 8

2.1. Availability of Healthcare Facilities 8

2.2. Authorisation of Healthcare Facilities 9

3. Status of Biomedical Waste Generation 11

3.1. Biomedical Waste Generation 11

3.2. Treatment of Biomedical waste 12

4. Status of Common Biomedical Waste Treatment Facilities (CBWTF) 14

4.1. Availability of CBWTF 14

4.2. Utilisation of CBWTF 15

4.3. Capacity of CBWTF 16

5. Status of Captive Treatment Facilities 17

6. Status of deep burial 20

6.1. Deep burial pits utilised by HCF 20

6.2. Deep burial pits utilised by CBWTF 20

7. Status of other essential compliances 22

7.1. Online Continuous Emission Monitoring System 22

7.2. Installation of Waste Water Treatment Facility 23

7.3. Implementation of Bar code and Unique Identification Number for HCF 23

8. Biomedical Waste Management in Healthcare Establishments of Armed Force Medical Services (AFMS) 24

9. Other issues of concern w.r.to bio-medical waste management 25

10. BMWM status in States and Union Territory 26

10.1. Andaman and Nicobar Islands 26

10.2. Andhra Pradesh 26

10.3. Arunachal Pradesh 27

10.4. Assam 27

10.5. Bihar 28

10.6. Chandigarh 28

10.7. Chhattisgarh 29

10.8. Daman &Diu and Dadra & Nagar Haveli 30
10.9. Delhi 30
10.10. Goa 31
10.11. Gujrat 31
10.12. Haryana 32
10.13. Himachal Pradesh 32
10.14. Jharkhand 33
10.15. Jammu and Kashmir 33
10.16. Karnataka 34
10.17. Kerala 35
10.18. Ladakh 35
10.19. Lakshadweep 36
10.20. Madhya Pradesh 36
10.21. Maharashtra 37
10.22. Manipur 37
10.23. Meghalaya 38
10.24. Mizoram 38
10.25. Nagaland 39
10.26. Odisha 39
10.27. Puducherry 40
10.28. Punjab 40
10.29. Rajasthan 41
10.30. Sikkim 42
10.31. Tamil Nadu 42
10.32. Telangana 43
10.33. Tripura 43
10.34. Uttarakhand 44
10.35. Uttar Pradesh 44
10.36. West Bengal 45

1. Introduction

1.1. Biomedical Waste Management Rules, 2016

Govt. of India has notified Biomedical Waste Management Rules, 2016 (BMWM Rules, 2016) under the Environment (Protection) Act, 1986 in the year 2016. These Rules stipulate various provision for collection, segregation, processing, treatment and disposal of bio-medical wastes in an environmentally sound manner.

As per the BMWM Rules, 2016 the biomedical waste generated in Healthcare Facilities should be collected, transported, treated and disposed-off through Common Biomedical Waste Treatment Facilities (CBWTFs). The Rules defines duties for Occupier which include hospitals, nursing homes, clinics, dispensaries, veterinary institutions, animal houses, pathological laboratories, etc.), duties of the operator of CBWTFs and duties for authorities like Ministry of Environment Forest & Climate Change (MoEF & CC), Ministry of Health & Family Welfare (MoH & FW), Ministry of Defence (MoD), Central Pollution Control Board (CPCB), State Government of Health (GoH), State Pollution Control Boards (SPCBs)/Pollution Control Committees (PCCs) and Municipalities or Urban Local Bodies (ULBs).

1.2. Annual Report on Biomedical Waste Management

As per BMWM Rules, 2016 every occupier or operator of common bio-medical waste treatment facility (CBWTF) shall submit an annual report to the prescribed authority i.e. SPCBs/PCCs on or before the 30th June of every year. Thereafter, the prescribed authority shall compile, review and analyse the information received and send to Central Pollution Control Board on or before the 31st July of every year. The BMWM Rules, 2016 Annual Reports shall also be available online on the websites of Occupiers, SPCB/PCC and CPCB. The occupier or operator of CBWTF through this report provide the information regarding the waste generated, collected in the facility, amount of biomedical waste treated, disposal facilities utilised, amount of treated biomedical waste sent to recyclers, number of vehicles used in collection of biomedical waste, trainings, accidents, violations of BMWM Rules, 2016, etc. These details help in analysing status of the biomedical waste across the country, identifying the gaps in the biomedical waste management and addressing the shortcoming in the upcoming year for effective management of biomedical waste and implementation of BMWM Rules, 2016.

1.3. Status of Annual Report 2022 and Gaps identified

All States, Union Territories and Director General of Armed Forces Medical Services (DGAFMS) have submitted the annual reports on biomedical waste management for the year 2022.

States and UTs namely Andaman & Nicobar, Arunachal Pradesh, Bihar, Chhattisgarh, Chandigarh, Delhi, DD & DNH, Goa, Gujarat, Himachal Pradesh, J&K, Karnataka, Kerala, Ladakh, Lakshadweep, Madhya Pradesh, Mizoram, Meghalaya, Nagaland, Odisha, Puducherry, Punjab, Sikkim, Telangana, Tamil Nadu, Rajasthan, Tripura, Uttarakhand and West Bengal & DGAFMS submitted the annual report information before July, 2023. However, SPCBs namely Andhra Pradesh, Assam, Jharkhand, Manipur, Maharashtra, Haryana, Uttar Pradesh submitted the annual report information after July, 2023.

The following common gaps/discrepancies were observed in the annual report information received from SPCBs/PCCs and communicated to respective States/UTs for clarification and rectification.

i) 11,093 HCFs are using the deep burial for disposal of biomedical waste.
ii) 24,705 HCFs are operational without authorisation.
iii) 17,490 HCFs are still using captive treatment facility.
iv) Cent percent installation of ETP by bedded HCFs is not achieved.
The response received from SPCBs/PCCs are incorporated in Annual Report.

1.4. Brief Summary of Bio-medical Waste Management in India

In year 2022, about 705 tonnes/day of biomedical waste was generated in the country, out of which 645 tonnes/day BMW was treated and disposed-off through CBWTFs and Captive treatment facilities. There are 3,93,939 no. of HCFs in the country, out of which 1,25,259 no. of HCFs are bedded with a total of 24,65,063 number of beds and 2,67,155 no. of HCFs are non-bedded. 3,10,809 no. of HCFs that is around 79% of HCFs utilise the facilities of CBWTFs for the collection, treatment, & disposal of biomedical waste, while 17,490 no. that is around 4.4 % of HCFs have captive treatment facilities for the treatment and disposal of biomedical waste. A gap of 16% of HCF is reported that are neither using CBWTF nor CTF. A gap of around 8.5% is reported between the generation and disposal of the BMW in the country, as per response received from SPCBs/PCCs the gap is due for information not received from CBWTFs or deep burials. There are 218 no. of CBWTFs in operation and 34 no. of CBWTFs

are under construction. Annexure I provide details of biomedical waste management in State/UTs Brief of biomedical waste scenario of the Country is given below:

No. of HCFs	**3,93,939**
No. of bedded HCFs	1,25,259
No. of non-bedded HCFs	2,67,155
No. of beds	24,65,063
No. of CBWTFs	218* + 34**
No. of HCFs granted authorization	1,56,637
No. of HCFs utilizing CBWTFs	3,10,809
No. of HCFs having Captive Treatment Facilities	17,490
No. of Captive Incinerators Operated by HCFs	153
Quantity of BMW generated in tons/day	705
Quantity of BMW treated in tons/day	645
No. of HCFs/CBWTFs violated BMW Rules	22,306
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	8291

Note: (i) * - CBWTFs in operation (ii) ** - CBWTFs under construction

1.5. Comparison of BMWM in 2021 and 2022

As per the annual report information for year 2022, 705 tons of BMW waste was generated in the country per day which is less than the generation of biomedical waste (764 tons per day) in year 2021. 59 tons (about 8%) of biomedical waste was generated less per day in 2022 in comparison to 2021 which may be due to increased quantity of biomedical waste during COVID-19 pandemic. Around 5% of increase in the total number of healthcare facilities have been reported in year 2022 as compared to 2021.

Annual report information shows that 3 new CBWTF were added in 2022 for the treatment & disposal of generated biomedical waste. Also, increase in number of HCFs utilising CBWTFs was observed to be increased from 2,62,786 to 3,10,809 from year 2021 and 2022, respectively. The captive treatment facilities are majorly operational in regions where CBWTFs is not accessible. Detailed comparison of biomedical waste management BMWM scenario during 2021 and 2022 is mentioned in Table 1 below:

Table 1: Comparative BMWM scenario in 2021 and 2022

Particulars	Year 2021	Year 2022
No. of HCFs	3,75,256	3,93,939
No. of bedded HCFs	1,21,396	1,25,259
No. of non-bedded HCFs	2,53,860	2,67,155
No. of beds	25,61,295	24,65,063
No. of CBWTFs	215* + 35**	218* + 34**
No. of HCFs utilizing CBWTF	2,62,786	3,10,809
No. of HCFs granted Authorization	3,20,751	1,56,637
No. of HCFs having Captive Treatment Facilities	13,605	17,490
No. of Captive Incinerators Operated by HCFs	102	153
Quantity of BMW generated (Tonnes/day)	764 (684 Non COVID BMW+80 COVID BMW)	705
Quantity of BMW treated (Tonnes/day)	721	645
No. of HCFs violated BMW Rules	23,199	22,306
No. of Show-cause notices/Directions issued todefaulter HCFs	15,355	8291

2. Status of Healthcare Facility

2.1. Availability of Healthcare Facilities

As per BMWM Rules, 2016, health care facility means a place where diagnosis, treatment or immunisation of human beings or animals is provided irrespective of type and size of health treatment system, and research activity pertaining thereto. It includes bedded and non-bedded HCFs.

As per the annual report information received from SPCBs and PCCs for the year 2022, there are 3,93,939 no. of HCFs, out of which 1,25,259 no. of HCFs are bedded and 2,67,155 no. of HCFs are non-bedded. Detail of number of HCFs in states/UTs and DGAFMS are given in fig. 1 and 2. In case of states, Maharashtra has the highest number of health care facilities followed by Karnataka and Uttar Pradesh. Sikkim has the lowest number of health care facilities followed by Arunachal Pradesh and Mizoram. In case of Union territories, Delhi has the highest number of health care facilities and the lowest healthcare facilities are present in Lakshadweep. Number of healthcare facilities generally depends on the geographical area of the state/UT, population density, availability of the health workers, financial restrains of state, workload of healthcare facility, etc. The figures also depict the same.

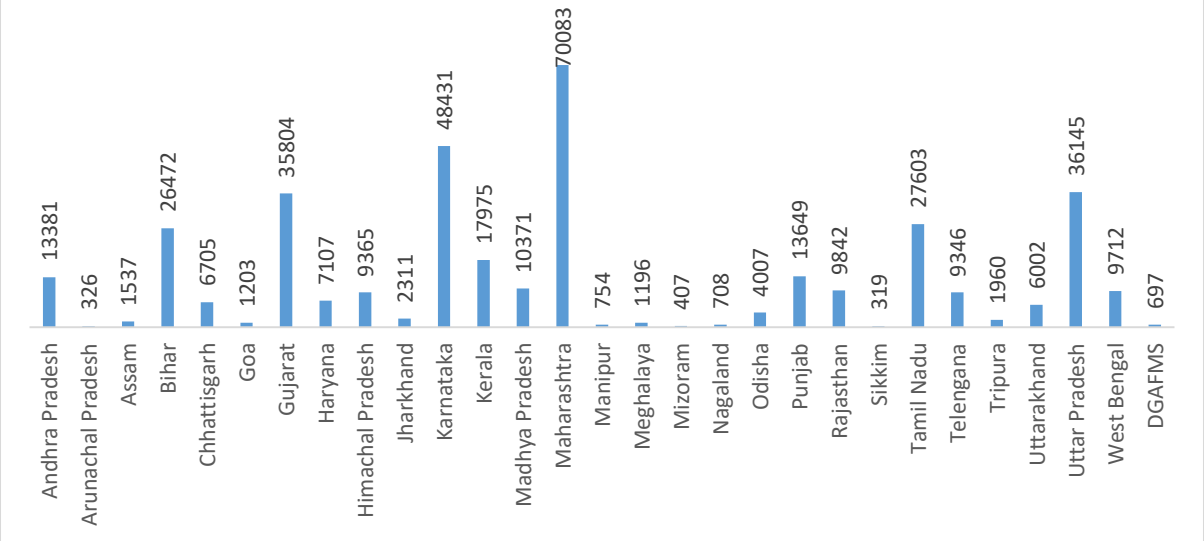


Figure 1: No. of HCFs in States & DGAFMS

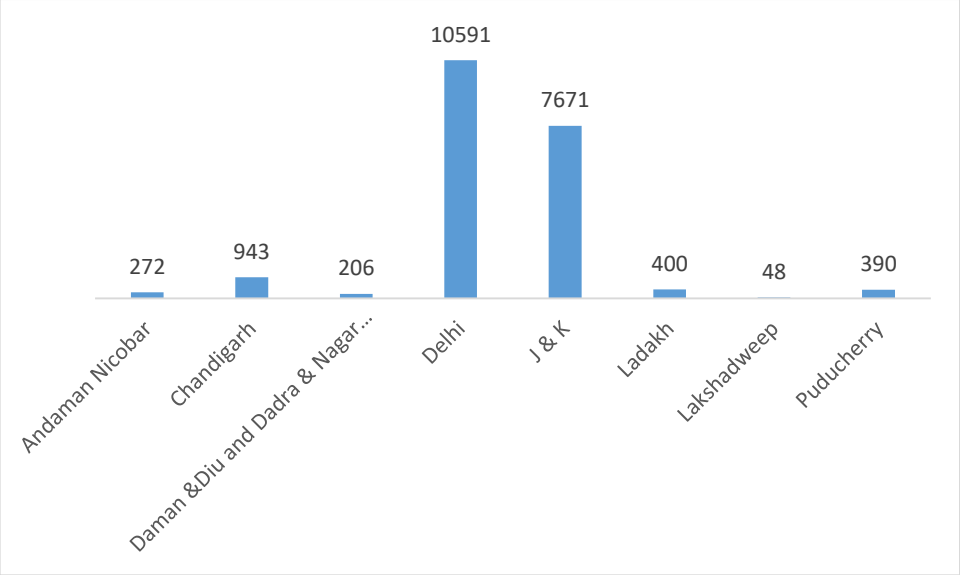


Figure 2: No. of HCFs in UT

2.2. Authorisation of Healthcare Facilities

Authorization is a process which brings HCFs and CBWTFs under the purview of BMWM Rules, 2016. BMWM Rules, 2016 stipulate that every healthcare facility (bedded & non bedded) require to obtain authorization from concerned SPCB/PCC. It has been observed that there are HCFs not yet authorized which is a violation of BMWM Rules, 2016. SPCB/PCC should ensure authorization of every HCFs under BMWM Rules, 2016.

HCFs of States/UTs namely Andhra Pradesh, Gujarat, Kerala, Lakshadweep, Manipur, Meghalaya, Mizoram, Nagaland, Rajasthan, Sikkim, Tripura and West Bengal have reported no HCF operating without authorisation. The States/UTs which has no HCF operating without authorisation are Delhi, Lakshadweep, Manipur, Mizoram, Nagaland and Puducherry, Tripura and West Bengal. As per information received from SPCBs/PCCs i) HCFs not applied for renewal of authorization after expiry of earlier authorization, ii) not applied for authorization after refusal of earlier authorization applications and iii) never applied for authorization.

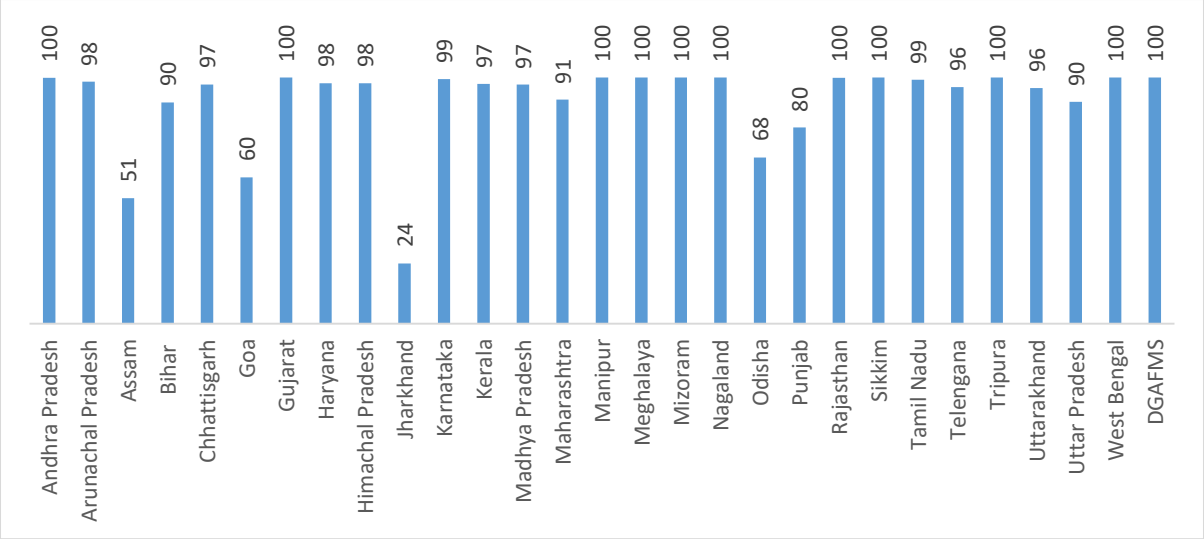


Figure 3: Percentage of applying HCF granted authorisation in the state and DGFAMS

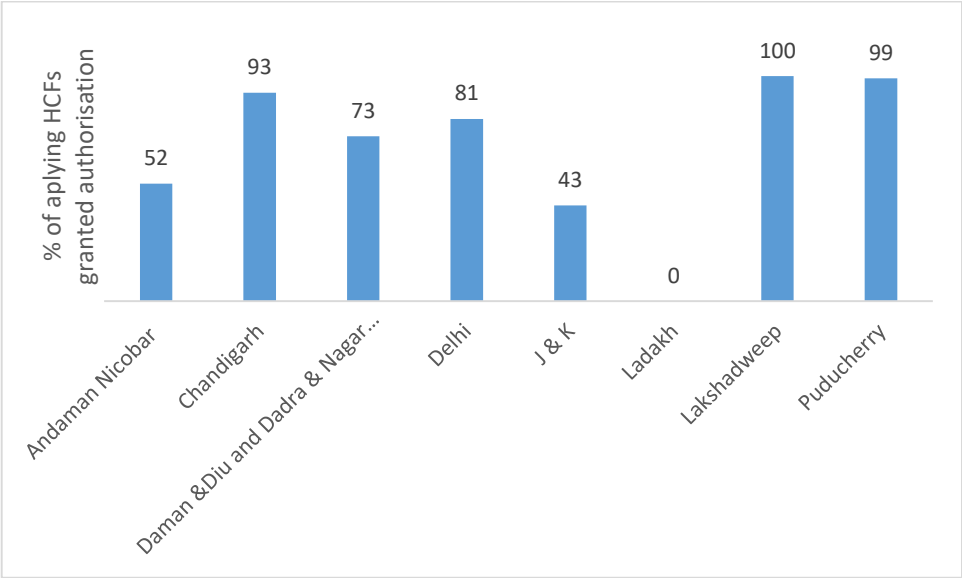


Figure 4: Percentage of applying HCFs granted authorisation in UTs

3. Status of Biomedical Waste Generation

3.1. Biomedical Waste Generation

Bio-medical waste means any waste, which is generated during the diagnosis, treatment or immunisation of human beings or animals or research activities pertaining thereto or in the production or testing of biological or in health camps, including the categories mentioned in Schedule I appended to BMWM Rules, 2016. As reported by SPCBs/PCCs, about 705 tonnes/day of BMW were generated during the year 2022 by 3,93,939 numbers of HCFs. The biomedical waste production generally depends on the number of healthcare facilities present in the state, population density, type of healthcare facility, etc. the quantity of biomedical waste produced in the states and UT in 2022 given in figure 5 and 6.

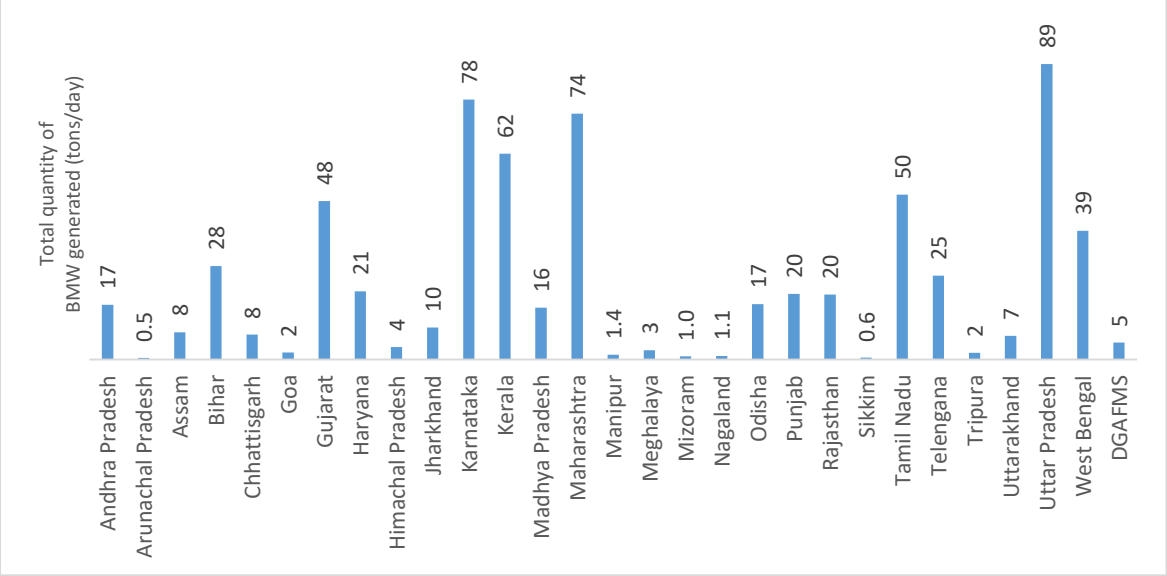


Figure 5: Total Quantity of BMW generated (tons/day) in different states and DGFAMS

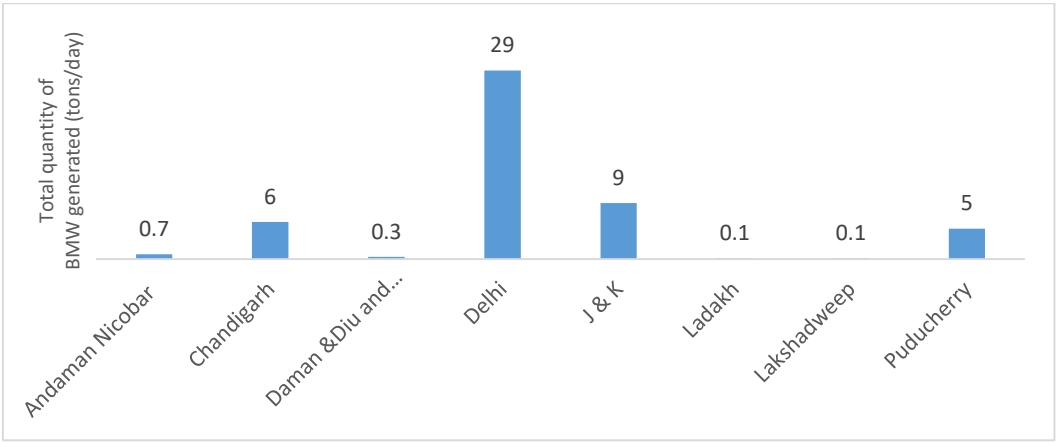


Figure 6: Total quantity of BMW generated (kg/day) in different UT

3.2. Treatment of Biomedical waste

As per Rule 7 (1), Bio-medical waste shall be treated and disposed of in accordance with Schedule I, and in compliance with the standards provided in Schedule-II by the health care facilities and CBWTFs. Out of 705 tonnes/day of BMW generated in the country in 2022, 645 tonnes/day of BMW is treated and disposed of by CBWTFs and captive treatment facilities (CTFs) installed by Healthcare Facilities.

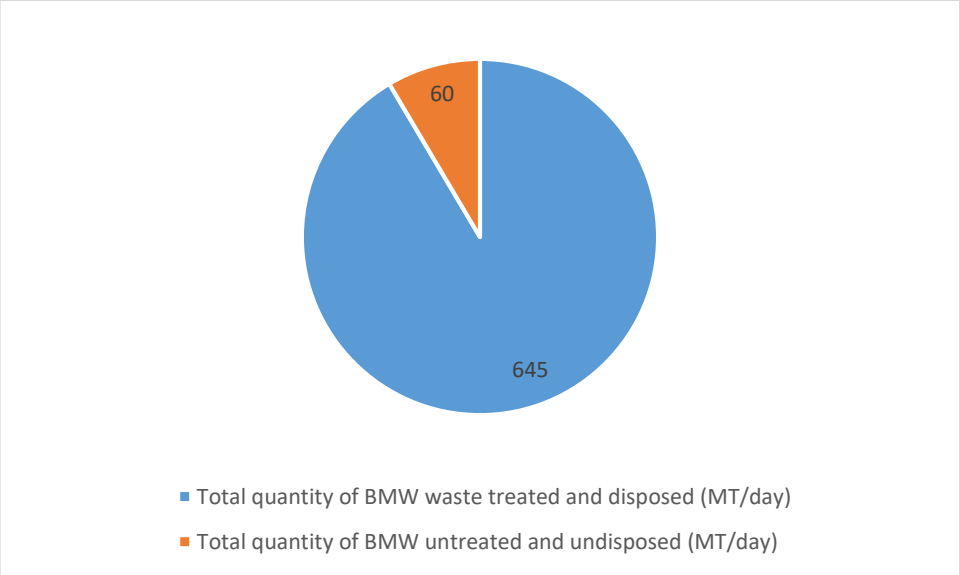


Figure 7: Quantity of BMW treated & disposed and gap in tonnes/day

As per Annual report information for year 2022, there is a gap between biomedical waste generation and its treatment & disposal. Figure 7 depicts the amount of BMW treated and disposed of, as well as the gap between treatment and disposal. For environment friendly disposal of BMW there should be no gap between generation and treatment & disposal of biomedical waste. However, the gap in generation, treatment & disposal has been observed in States namely Assam, Bihar, Jharkhand, Karnataka, Kerala, Madhya Pradesh, Meghalaya and Uttarakhand (fig. 8). Among the above States, the gap in generation, treatment, and disposal has also been observed in 2021 for Assam, Bihar, Karnataka, Jharkhand, Karnataka, Madhya Pradesh while the states of Himachal Pradesh, Nagaland and Tripura has overcome the gap in 2022 for the generation and treatment of BMW. CPCB has communicated the gaps observed to the respective SPCB/PCC to rectify the issue and ensure the disposal of all BMW generated in accordance with the BMWM Rules, 2016.

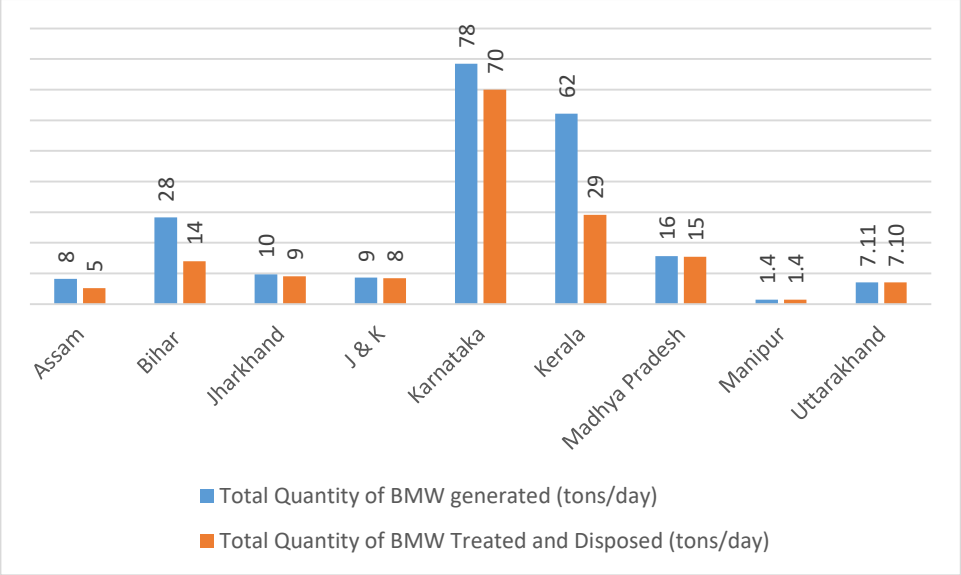


Figure 8: Status of generation, treatment & disposal of BMW (tons/day)

Some SPCBs/PCCs submitted that the gap is due to incomplete information received from HCFs and CBWTFs or deep burials used for biomedical waste.

4. Status of Common Biomedical Waste Treatment Facilities (CBWTF)

4.1. Availability of CBWTF

CBWTF means any facility which is involved in collection, treatment and disposal of bio-medical waste. As per BMWM Rules, 2016, it is the duty of CBWTF operator to take all necessary steps to ensure that the BMW collected from the occupier is transported, handled, stored, treated and disposed of, without any adverse effect to the human health and the environment, in accordance with BMWM Rules, 2016 and guidelines issued by the CPCB.

As per the Annual Report Information submitted by SPCBs/PCCs for the year 2022, currently there are 218 numbers of CBWTFs operated in the Country and 34 CBWTFs are under construction. In year 2022, there is increase in number of CBWTFs in Assam, Chhattisgarh, Punjab, Tripura and West Bengal as compared to year 2021. Andaman and Nicobar, Arunachal Pradesh, Laddakh, Lakshadweep, Mizoram, Nagaland and Sikkim have reported that no CBWTF is present in the State/UTs. Daman & Diu and Dadra & Nagar Haveli utilises the CBWTF of Gujarat state for treatment of biomedical waste. Similarly, Lakshadweep is utilising the facility of CBWTF situated in Kerala. In rest reported states the disposal of Biomedical waste is undertaken using the Captive treatment facility or deep burials. In this regard, CPCB requested such SPCBs/PCCs for submitting the proposal for setting up of CBWTFs for which they may avail financial assistance from Ministry of Environment Forests & Climate Change. No. of CBWTF present in state and UTs is represented in fig. 9 and fig. 10.

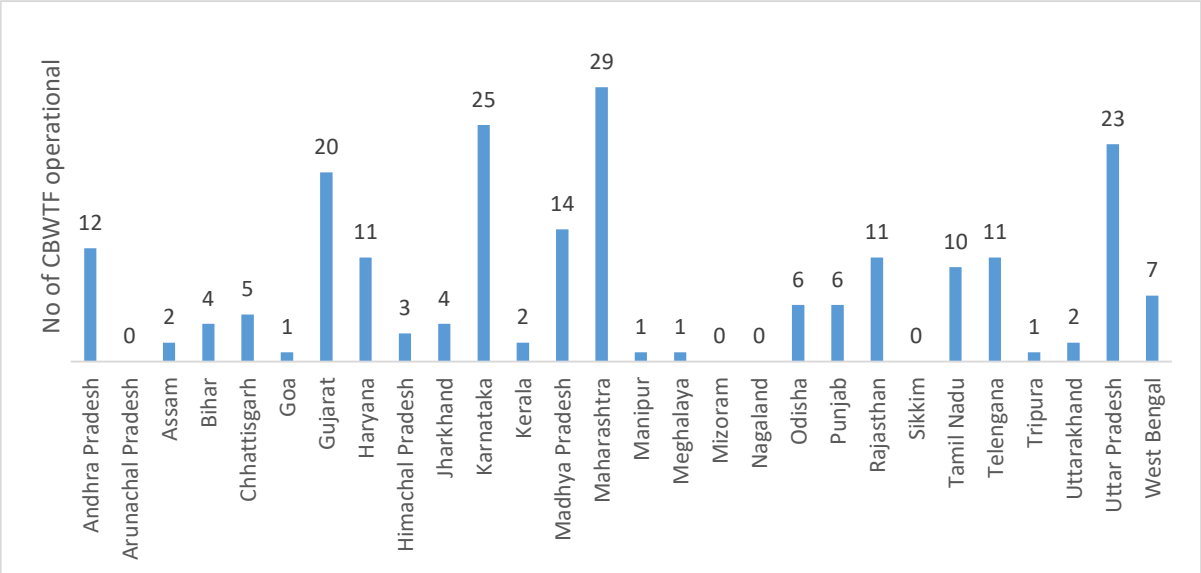


Figure 9: No. of CBWTFs operational in the States

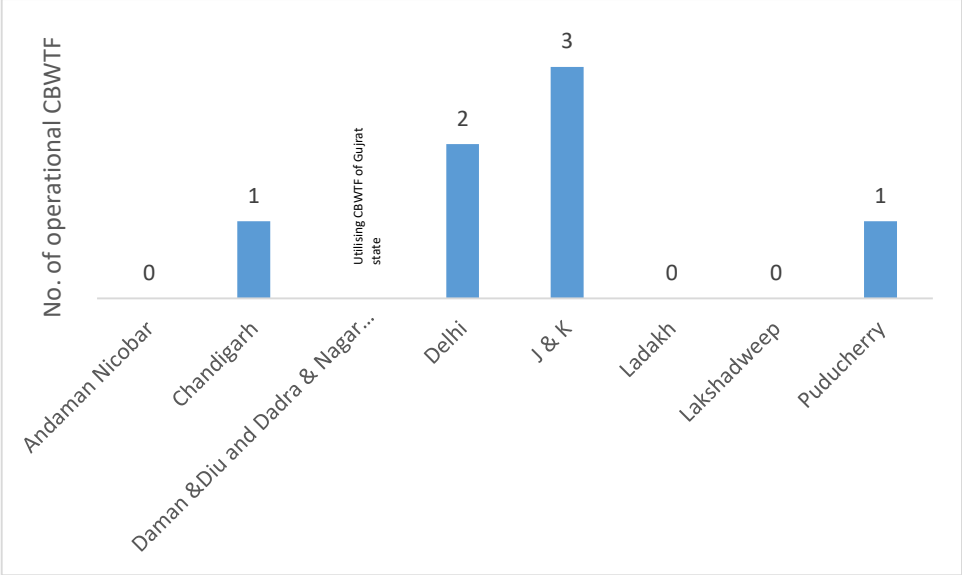


Figure 10: Number of CBWTF operational in UTs

As shown in figure 7 and 8, the availability of CBWTFs operated in States varies based on the area of the State, population density, number of HCFs in the states/UT, capacity of the CBWTF, accessibility of CBWTF, etc.

New CBWTF are under currently reported to be under construction in the states of Andhra Pradesh, Chhattisgarh, Gujarat, Himachal Pradesh, Jharkhand, Karnataka, Madhya Pradesh, Maharashtra, Rajasthan, Tamil Nadu, Uttar Pradesh, Uttarakhand and West Bengal.

4.2. Utilisation of CBWTF

As per Rule 7 (3) of the BMWN rules, 2016, "no occupier shall establish on-site treatment and disposal facility, if a service of common biomedical waste treatment facility is available at a distance of seventy-five kilometre". As per Annual Report information for year 2022, all HCFs operational in States/UTs namely Chandigarh, Delhi, Gujrat, Haryana, Puducherry, Tamil Nadu, Telangana and West Bengal are using CBWTF for treatment and disposal of biomedical waste. Less than 30 percent of HCFs are utilising CBWTF in the States/UT namely Meghalaya, Tripura, Assam and Bihar and is depended largely on the captive treatment and deep burials for treatment and disposal of biomedical waste. The reason of not using CBWTF is may be non-accessibility of CBWTF due to hilly terrain and presence of the HCF in the remote locations. States/UTs may look into the accessibility and come up with new CBWTFs to cover all HCFs in the State/UT. Utilization of CBWTFs by HCFs (in percentage) is given in fig. 11.

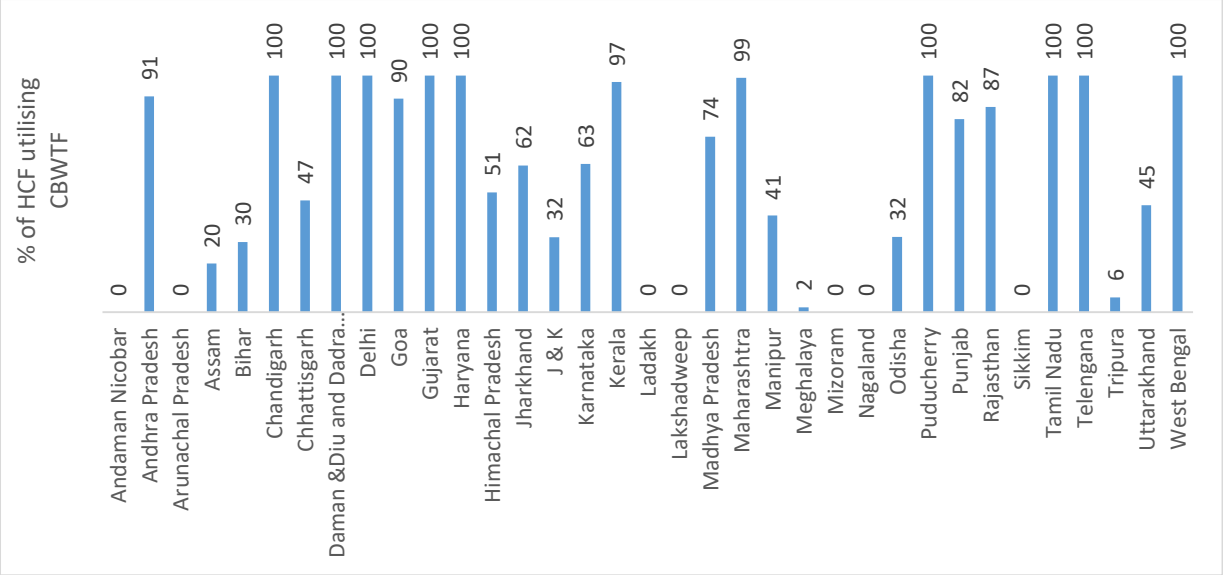


Figure 11: Percentage of HCF utilising the CBWTF in various states/UT

4.3. Capacity of CBWTF

The CBWTFs in the country are operating at cumulative treatment and disposal capacity of 1590 MT/day, of which incineration capacity is 858 MT/day and autoclave capacity is 732 MT/day. The present generation of 705 MT/day may look adequate for treatment and disposal of BMW. However, availability of CBWTFs may vary at State Level. Available treatment capacity of CBWTFs and utilisation of capacity in the country is shown in fig. 12.

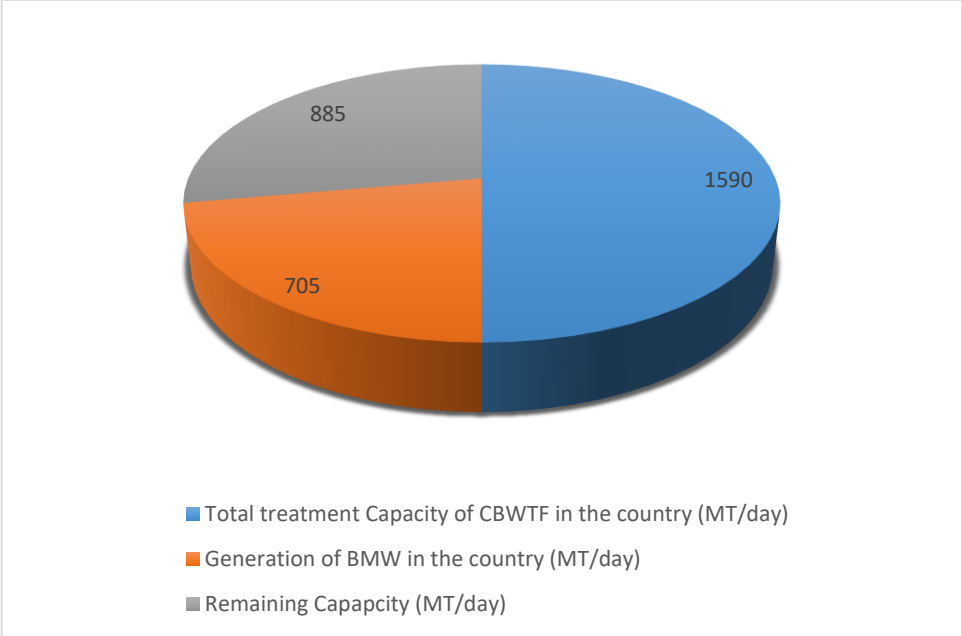


Figure 12: Treatment capacity available and utilization of capacity of CBWTFs in the country

5. Status of Captive Treatment Facilities

BMWM Rules, 2016 restricts on-site treatment and disposal facility, if a service of CBWTF is available at a distance of 75 Km. In cases where service of the common bio-medical waste treatment facility is not available, the Occupiers shall set up requisite biomedical waste treatment equipment like incinerator, autoclave or microwave, shredder prior to commencement of its operation, as per the authorisation given by the prescribed authority.

As per the annual report information submitted by SPCBs/PCCs, in 2022, 17,490 number of Healthcare Facilities in the country are utilising the captive treatment facility. Comparing to 2021, an increase in the number of captive treatment facilities operated by HCFs is reported in 2022 (fig. 13). Further, CPCB has also issued directions to all SPCBs/PCCs regarding to ensure compliance to the standards prescribed under BMWM Rules, 2016 by captive treatment facilities operated by HCFs.

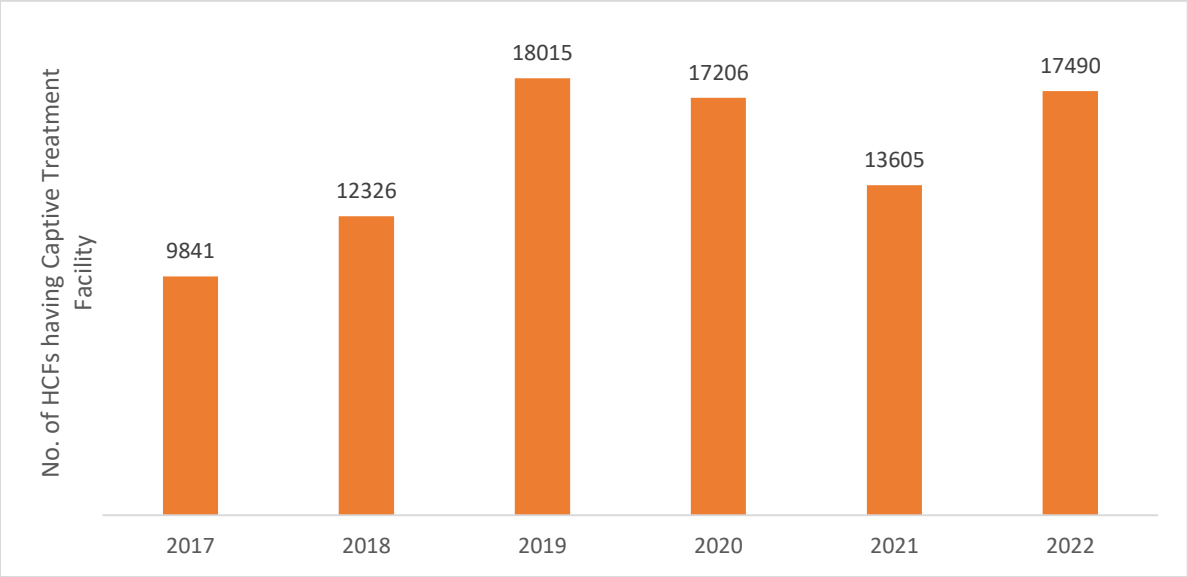


Figure 13: Number of HCFs having captive treatment facilities

HCFs of the States/UT namely Andaman and Nicobar, Arunachal Pradesh, Assam, Bihar, Chhattisgarh, Himachal Pradesh, Jharkhand, Karnataka, Kerala, Ladakh, Lakshadweep, Madhya Pradesh, Manipur, Meghalaya, Mizoram, Nagaland, Puducherry, Sikkim, Tripura, Uttarakhand and Uttar Pradesh are operating captive treatment facility (Fig. 14 & Fig. 15). Himachal Pradesh has reported to be using highest number of captive treatment facilities that is around 50% of the HCFs present in the states are using the CTF. In year 2022, no CTF are being utilised in the state of Andhra Pradesh, Goa, Gujrat Haryana, Punjab, Tamil Nadu, Telangana and West Bengal. In UTs, Ladakh has the highest number of HCF utilising the CTF.

The UT of Chandigarh, Daman and Diu and Dadar and Nagar Haveli and Jammu & Kashmir is having no CTF utilised by the HCF. The percentage of HCF out of total HCFs present in the states/UTs utilising the CTF are given in figure 16. 100% of the waste in Arunachal Pradesh, Mizoram and Lakshadweep is being disposed-off the CTF. These States/UT Should explore the possibility of setting up of CBWTF.

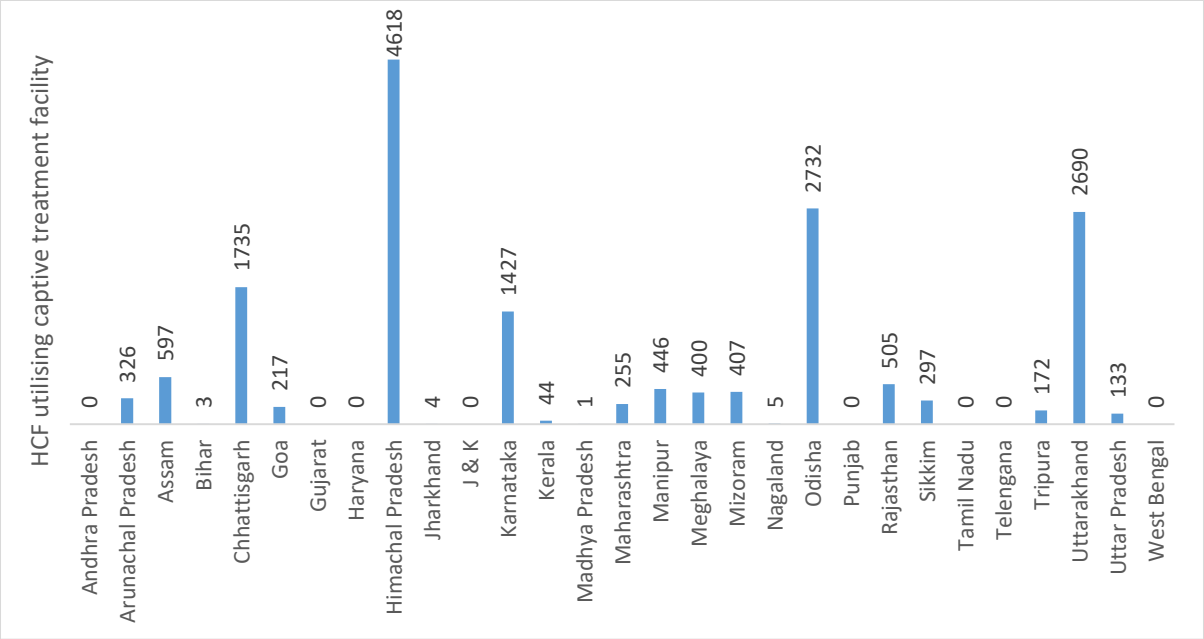


Figure 14: Number of HCF having captive treatment in different states

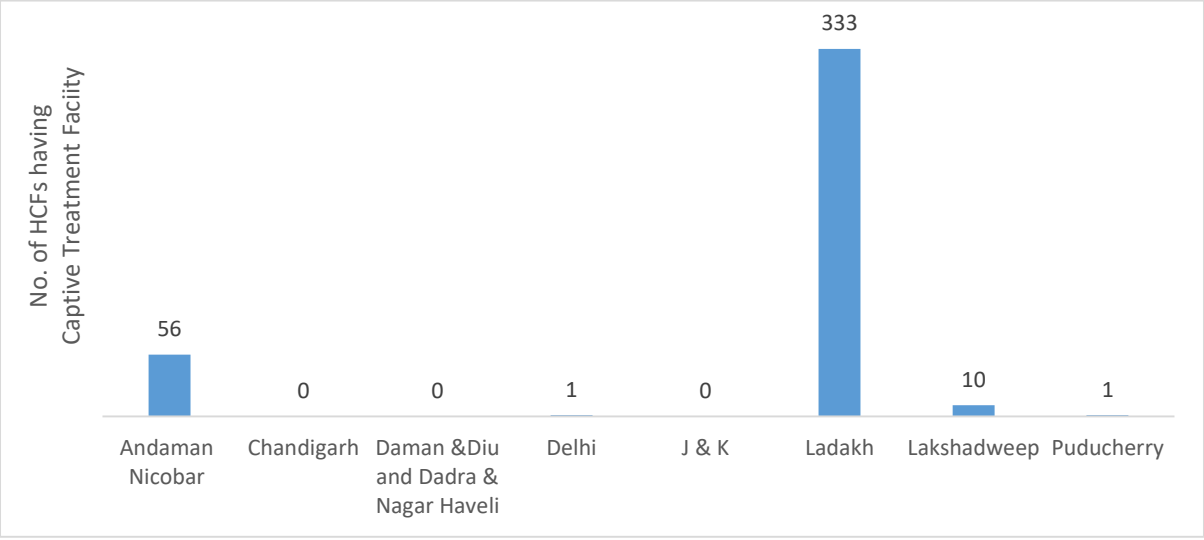


Figure 15: Number of HCF having captive treatment facility in different UTs

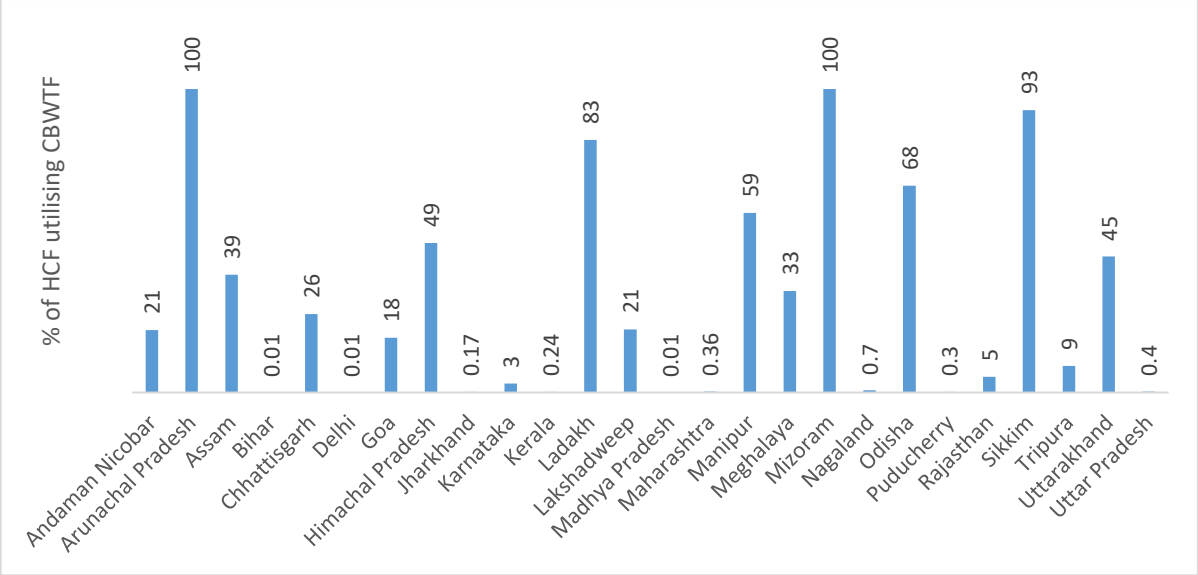


Figure 16: % of HCF utilising the CTF in various states and UT

6. Status of deep burial

6.1. Deep burial pits utilised by HCF

As per BMWM rules, 2016 disposal by deep burial is permitted only in rural or remote areas where there is no access to common bio-medical waste treatment facility. This will be carried out with prior approval from the prescribed authority and as per the Standards specified in Schedule-III. The deep burial facility shall be located as per the provisions and guidelines issued by Central Pollution Control Board from time to time. As per the information submitted by the SPCB/PCCs, deep burial pits installed by the HCFs of Andaman & Nicobar, Arunachal Pradesh, Himachal Pradesh, Jharkhand, Kerala, Ladakh, Lakshadweep, Maharashtra, Meghalaya, Mizoram, Nagaland, Odisha, Puducherry, Rajasthan, Sikkim and Tripura. The status of deep burial pits installed by HCFs is given in fig. 17. Highest number of HCFs in Himachal Pradesh are using deep burial pits for disposal of BMW. But the numbers have decreased to 4618 in 2022 from 5174 in 2021. UT of Puducherry has lowest that is only one HCF using the deep burial for disposal of BMW.

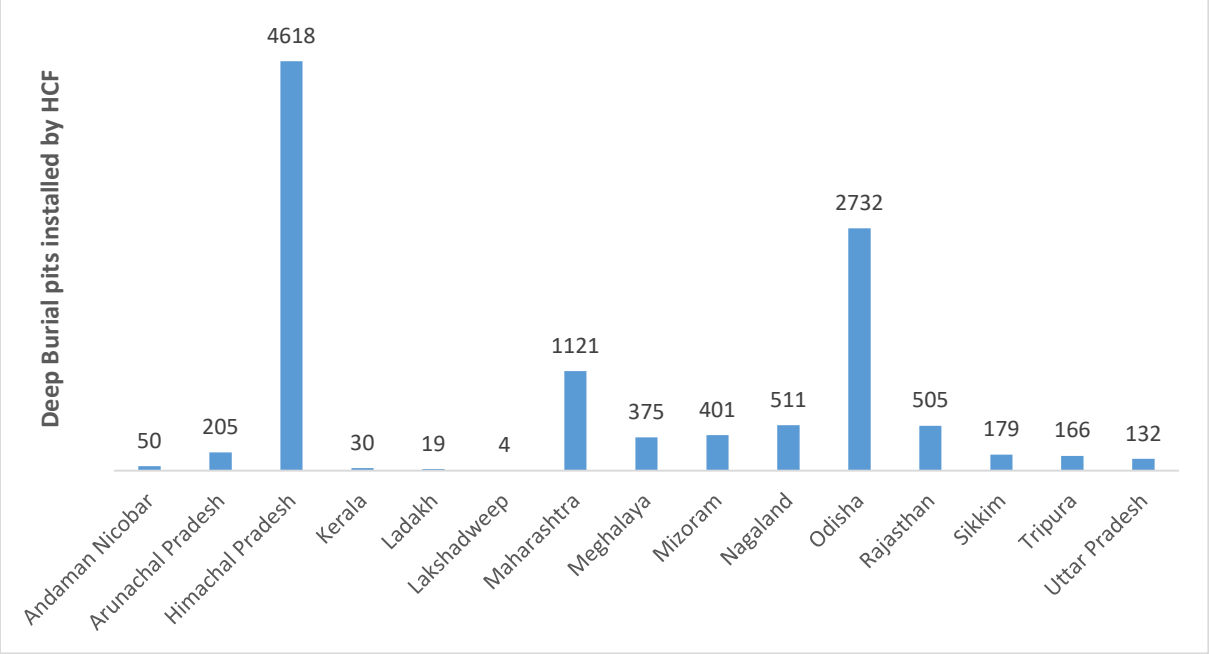


Figure 17: Deep burial installed by HCFs in States/UTs

6.2. Deep burial pits utilised by CBWTF

As per the revised guidelines for CBWTFs, SPCB/PCC should not allowed deep burial of BMW as a part of CBWTF. Any existing CBWTF having disposal of BMW by deep burial should have the requisite treatment equipment as stipulated under the BMWM Rules. CPCB issued directions to all SPCBs/PCCs regarding verification of deep burial pits that they are

authorized and designed in line with standards given under BMWM Rules, 2016. As per the response received by the Odisha PCB, the 2 deep burial pits though existing, are currently not in use. Response from other state boards are still awaited. Respective State Boards should initiate steps to set up CBWTF so as to avoid usage of deep burial pits.

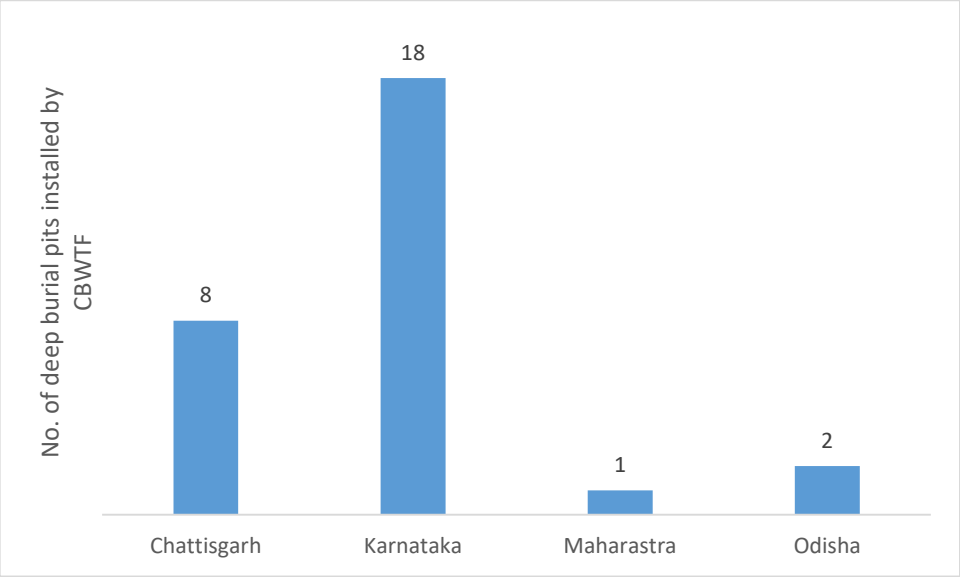


Figure 18: No. of deep burial pits installed by CBWTF

7. Status of other essential compliances

7.1. Online Continuous Emission Monitoring System

BMWM Rules, 2016 stipulates that every occupier or operator of the common bio-medical waste treatment facility shall install Online Continuous Emission Monitoring System (OCEMS) for the parameters as stipulated by SPCBs/PCCs in authorization and transmit the real time data to the servers at SPCB/PCC and CPCB. As per the submitted information, 208 out of 218 nos. that is about 95% of present CBWTFs have installed OCEMS with their incinerators. Further, it has been observed by CPCB that some of the CBWTFs connected to OCEMS portal are not transmitting data regularly to CPCB server. CPCB has conducted meetings with CBWTF operators related to connectivity and calibration of OCEMS and also issued directions to all SPCBs/PCCs regarding to ensure OCEMS data transmission to CPCB server. Details of CBWTFs installed OCEMS is given in fig. 19.

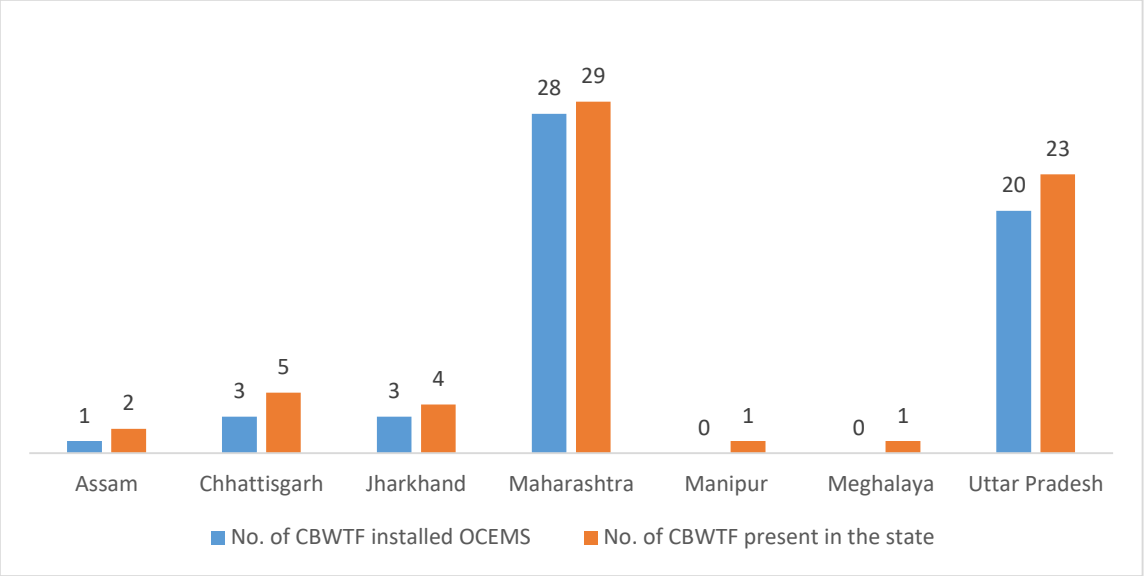


Figure 19: Number of CBWTF installed OCEMS out of total CBWTF present in the State/UT

All operational CBWTFs of States namely Andhra Pradesh, Bihar, Chandigarh, Delhi, Goa, Haryana, Himachal Pradesh, J & K, Karnataka, Kerala, Madhya Pradesh, Odisha, Puducherry, Punjab, Rajasthan, Telangana, Tamil Nadu, Uttarakhand and West Bengal have installed OCEMS, however few operational CBWTFs of states namely Assam, Chhattisgarh, Jharkhand, Maharashtra, Manipur and Meghalaya have not installed OCEMS.

7.2. Installation of Waste Water Treatment Facility

As per Schedule I of BMWM Rules, 2016, the chemical liquid waste shall be pre-treated before mixing with other wastewater. The combined discharge shall confirm to the discharge norms given in Schedule III. Further, bedded HCFs are required to comply with the standard prescribed for liquid waste under Schedule II of BMWM Rules, 2016. However, HCFs may not necessarily need to install Effluent Treatment Plant (ETP) in case waste water is discharged into a public sewer connected to a terminal sewage treatment plant (STP). However, it should meet the desired general standards as notified under the Environment (Protection) Act, 1986. In terms of installing of Waste Water Treatment Facility, several States are not up to par.

7.3. Implementation of Bar code and Unique Identification Number for HCF

As per BMWM Rules, 2016, it is the duty of every occupier and operator of CBWTF to adopt bar code system for tracking of bio-medical waste. Also, as per order of Hon'ble Supreme Court of India in the matter of IA No. 181745 of 2019 and I.A. No. 46339 of 2020, in W.P.C. 13029 of 1985 directed MoEF&CC / CPCB to work out a national bar-coding system/portal for tracking of biomedical waste. In this regard, CPCB has initiated the project namely "Implementation of Centralised Bar Code System for Tracking Biomedical Waste (CBST-BMW)" and finalised the vendor M/s Atishay Limited, Bhopal for the same. This project will help in tracking of the BMW from source of generation to the final disposal. CPCB has also directed the SPCBs/PCCs to develop a system to allot unique identity number to all the HCFs present in the states/UTs. CPCB has directed all SPCBs to timely finish the implementation of the Bar code and Unique Identification Number for HCF.

8. Biomedical Waste Management in Healthcare Establishments of Armed Force Medical Services (AFMS)

The prescribed authority for Armed Force Medical Services is DGAFMS. As reported, 697 HCFs are established under DGAFMS that generates about 5.1 tonnes/day bio-medical waste, which is entirely treated and disposed-off by CBWTFs and Captive Treatment Facility. Out of 697 no. of HCFs, there are 226 bedded HCFs and Nursing homes, 465 Clinics, Dispensaries, 2 Pathological laboratories, 2 Blood banks and 2 Research Institutions. Out of 697 no. of HCFs, 622 no. of HCFs are using CBWTFs and 75 no. of HCFs are using Captive Treatment Facility. Further, 0.9 tonnes/day of BMW is being treated by captive treatment facilities of HCFs, which is not recommended under BMWM Rules, 2016 and 4.2 tonnes/day BMW is being treated and disposed-off by CBWTF. As per the clarification received 75 HCF are located in remote location/isolated far flung areas where CBWTF is not available within the range of 75 km. However, AFHCEs is making a continuous effort to reduce the number of HCFs disposing through captive treatment facility. The number of HCFs disposing through captive treatment facility have been reduced to 75 in 2022 as compared to 82 in 2021. 23 AFHCEs located in Assam, Bihar, Goa, Odisha, Kerala, Kolkata, Karnataka, Maharashtra, Madhya Pradesh, Rajasthan and Uttar Pradesh were inspected by the CPCB and the non-compliance were observed in 10 AFHCEs which were communicated to the concerned authority along with recommendations. CPCB also conducted a meeting on 04.07.2023 with officials of DGAFMS to discuss the implementation of biomedical waste management Rules, 2016 by AFHCFs. The shortcomings observed during inspection were discussed with the officials of DGAFMS and following decisions were taken:

- Time bound Action plan may be submitted to correct shortcomings observed during inspection.
- Regular meetings of Advisory Committee (Defence) may be conducted to resolve the issues and gaps in implementation of BMWM Rules, 2016.
- The matter of installation of ETP may be discussed with Advisory Committee (Defence).
- DGAFMS may organize a workshop for awareness on Biomedical Waste Management in line with the BMWM Rules, 2016.
- Separate modules may be developed in the project “Implementation of Centralised Barcode System for Tracking of Biomedical Waste” for AFHCEs.

9. Other issues of concern w.r.to bio-medical waste management

- 100% HCFs are still not authorisation even after 7 years of the notification of the BMWM, 2016.
- HCFs are using captive waste treatment facilities in some states instead of CBWTF, SPCB are directed to undertake continuous monitoring of such facilities.
- There are less number of CBWTF present in some states in comparison to the population density and number of HCF present. States are required to increase the numbers of CBWTF present so more HCF can utilise CBWTF in place of captive treatment facility.
- District Level information of BMWM is not available in every State/UT which is required as per the BMWM Rules, 2016 as well as CPCB guidelines.
- Domestic biomedical waste is not collected separately from the households by the Urban Local Bodies as required under BMWM Rules, 2016.
- Some CBWTFs and HCFs in the States/UTs have installed deep burial and many of the States/UTs has not provided information w.r.to deep burial pits installed by HCFs. Status of compliance by deep burial should be verified by SPCBs/PCCs.
- Liquid waste treatment facility i.e ETP/STP is not installed by all bedded HCFs in the States.
- Gap analysis to assess the requirement of additional treatment facility to treat and dispose of the biomedical waste is not conducted as per CPCB guidelines.
- Every CBWTFs and captive treatment facility should install OCEMS and ensure its connectivity to SPCB/PCC and CPCB server.

10. BMWM status in States and Union Territory

10.1. Andaman and Nicobar Islands

No. of HCFs	272
No. of bedded HCFs	53
No. of non-bedded HCFs	219
No. of beds	1610
No. of CBWTFs	NIL
No. of HCFs granted authorization	128
No. of HCFs utilizing CBWTFs	NIL
No. of HCFs having Captive Treatment Facilities	6
No. of Captive Incinerators Operated by HCFs	6
Quantity of BMW generated in kg/day	734.79
Quantity of BMW treated in kg/day	734.79
No. of HCFs/CBWTFs violated BMW Rules	NIL
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	130

Andaman and Nicobar Islands are geographically isolated and most HCFs are located in the remote locations, therefore deep burial pits are being used by the HCF for disposal of biomedical waste. Currently there is no operational CBWTF in the UT, although Directorate of Health Service (DHS) has been identifying land for installation of CBWTF. All generated BMW that is 734.79 kg is being treated and disposed through captive incinerators and deep burial pits. CPCB strongly recommend commissioning of CBWTF in the UT so that dependency of HCF on captive treatment and deep burial can be reduced.

10.2. Andhra Pradesh

No. of HCFs	13,381
No. of bedded HCFs	6,779
No. of non-bedded HCFs	6,602
No. of beds	1,62,751
No. of CBWTFs	12
No. of HCFs granted authorization	3074
No. of HCFs utilizing CBWTFs	13381
No. of HCFs having Captive Treatment Facilities	NIL
No. of Captive Incinerators Operated by HCFs	NIL
Quantity of BMW generated in kg/day	16,512.02
Quantity of BMW treated in kg/day	16,512.02
No. of HCFs/CBWTFs violated BMW Rules	907
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	907

Currently all the BMW generated in the state is treated through CBWTF and no captive treatment facility is operational in the state. There are 781 occupiers that are currently operational without authorisation in the state. Only 615 occupier installed liquid waste treatment facility. SPCB should authorise the remaining occupiers and improve the compliance regarding liquid waste treatment.

10.3. Arunachal Pradesh

No. of HCFs	326
No. of bedded HCFs	189
No. of non-bedded HCFs	137
No. of beds	2759
No. of CBWTFs	NIL
No. of HCFs granted authorization	403
No. of HCFs utilizing CBWTFs	NIL
No. of HCFs having Captive Treatment Facilities	326
No. of Captive Incinerators Operated by HCFs	16
Quantity of BMW generated in kg/day	464.6
Quantity of BMW treated in kg/day	464.6
No. of HCFs/CBWTFs violated BMW Rules	56
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	56

Currently there is no CBWTF operational in the state and the generated biomedical that is 464.6 kg/day is treated and disposed through captive treatment facility. Though the HCFs are located in remote areas, development of CBWTF should be explored by SPCB to avoid use of captive treatment facility and deep burial.

10.4. Assam

No. of HCFs	1537
No. of bedded HCFs	767
No. of non-bedded HCFs	770
No. of beds	30,432
No. of CBWTFs	02
No. of HCFs granted authorization	516
No. of HCFs utilizing CBWTFs	315
No. of HCFs having Captive Treatment Facilities	597
No. of Captive Incinerators Operated by HCFs	28
Quantity of BMW generated in kg/day	8229.3
Quantity of BMW treated in kg/day	5208.9
No. of HCFs/CBWTFs violated BMW Rules	482

No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	81

Currently there are 02 CBWTF is present in the state and 2,140.5 kg/day of BMW is treated and disposed through CBWTF. 597 of HCF in the state have captive treatment facility and 3,068.5 kg/day of BMW is treated and disposed through CTF. A gap of around 37% is observed between the generation and disposal of the BMW in the state. 526 occupiers are currently in operation without applying for authorisation. 778 nos. of occupier have installed the liquid waste treatment facility. SPCB should ensure implementation of BMWM Rules, 2016 and bridge the gaps identified in Annual Report

10.5. Bihar

No. of HCFs	26,472
No. of bedded HCFs	6,061
No. of non-bedded HCFs	20,411
No. of beds	1,04,943
No. of CBWTFs	04
No. of HCFs granted authorization	2311
No. of HCFs utilizing CBWTFs	7836
No. of HCFs having Captive Treatment Facilities	03
No. of Captive Incinerators Operated by HCFs	03
Quantity of BMW generated in kg/day	28246.3
Quantity of BMW treated in kg/day	13949.3
No. of HCFs/CBWTFs violated BMW Rules	2542
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	29

Currently there are only 3 CBWTF are operational in the state which seems less to cater the need of the state. There is a gap of around 50% between the generated and treated BMW in the state. 18633 HCF facilities are neither having the captive treatment nor utilising the CBWTF. 2811 HCF are operational in the state without authorisation. 778 nos. of occupier have installed the liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report.

10.6. Chandigarh

No. of HCFs	943
No. of bedded HCFs	44
No. of non-bedded HCFs	899
No. of beds	4825
No. of CBWTFs	01
No. of HCFs granted authorization	50

No. of HCFs utilizing CBWTFs	943
No. of HCFs having Captive Treatment Facilities	0
No. of Captive Incinerators Operated by HCFs	0
Quantity of BMW generated in kg/day	5710
Quantity of BMW treated in kg/day	5710
No. of HCFs/CBWTFs violated BMW Rules	0
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	0

Currently there is 01 CBWTF is present in the UT and total generated BMW that is 5710 kg/day is treated and disposed through CBWTF. There is no captive treatment facility in the UT and all occupiers are currently authorised by PCC. 899 non bedded facilities pre-treat their liquid waste, neutralize it and then dispose it off in the drains whereas 15 nos. out 44 nos. of bedded HCF have the ETP and rest are in the installation phase. ETP should be established by remaining HCF for proper disposal of liquid waste.

10.7. Chhattisgarh

No. of HCFs	6,705
No. of bedded HCFs	2,292
No. of non-bedded HCFs	4,280
No. of beds	50,829
No. of CBWTFs	5
No. of HCFs granted authorization	4,513
No. of HCFs utilizing CBWTFs	3168
No. of HCFs having Captive Treatment Facilities	1735
No. of Captive Incinerators Operated by HCFs	2
Quantity of BMW generated in kg/day	7504.9
Quantity of BMW treated in kg/day	8210.8*
No. of HCFs/CBWTFs violated BMW Rules	196
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	134

*As the no. of HCFs which have submitted the annual report is much less than the no. of HCFs which have been given authorization, the quantity of Bio Medical Waste treated is more than the quantity generated.

Currently there are 5 CBWTF operational in the state of which 3 are incinerator based and 2 are deep burial pits based. 3 CBWTF are currently under construction in the state. The deep burial pits based CBWTF are allowed until the new incinerator based CBWTFs are established, after which their operation will be terminated. 1225.78 kg/day of BMW is treated and disposed through captive treatment facility. 1287 occupier have installed liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report.

10.8. Daman &Diu and Dadra & Nagar Haveli

No. of HCFs	206
No. of bedded HCFs	74
No. of non-bedded HCFs	132
No. of beds	1,449
No. of CBWTFs	Utilising CBWTF of Gujarat
No. of HCFs granted authorization	79
No. of HCFs utilizing CBWTFs	206
No. of HCFs having Captive Treatment Facilities	0
No. of Captive Incinerators Operated by HCFs	0
Quantity of BMW generated in kg/day	324
Quantity of BMW treated in kg/day	324
No. of HCFs/CBWTFs violated BMW Rules	2
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	2

Currently only 1 CBWTF which is situated in the Gujarat state is utilised by the HCF of the UT for disposal of the BMW. No captive treatment facility is operational in the UT. No occupier in the UT has installed liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report.

10.9. Delhi

No. of HCFs	10,591
No. of bedded HCFs	1,141
No. of non-bedded HCFs	9,450
No. of beds	60,344
No. of CBWTFs	2
No. of HCFs granted authorization	1,253
No. of HCFs utilizing CBWTFs	10,591
No. of HCFs having Captive Treatment Facilities	1
No. of Captive Incinerators Operated by HCFs	0
Quantity of BMW generated in kg/day	29,155
Quantity of BMW treated in kg/day	29,155
No. of HCFs/CBWTFs violated BMW Rules	25
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	25

Currently there are 02 CBWTF is present in the UT and 29,155 kg/day of BMW is treated and disposed through CBWTF. 01 HCF in the UT have captive treatment facility and 402.2 kg/day of BMW is treated and disposed through CTF. No occupier is currently in operation without

applying for authorisation. 157 nos. of occupier have installed the liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report.

10.10. Goa

No. of HCFs	1,203
No. of bedded HCFs	168
No. of non-bedded HCFs	1,035
No. of beds	6,573
No. of CBWTFs	1
No. of HCFs granted authorization	241
No. of HCFs utilizing CBWTFs	1,086
No. of HCFs having Captive Treatment Facilities	0
No. of Captive Incinerators Operated by HCFs	0
Quantity of BMW generated in kg/day	2,128.4
Quantity of BMW treated in kg/day	2,128.4
No. of HCFs/CBWTFs violated BMW Rules	336
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	0

Currently there is 01 CBWTF is present in the state and total generated waste that is 2128.4 kg/day is treated and disposed through CBWTF. There is no captive treatment facility in the state. For the year 2022, the GSPCB is in process of identifying additional HCFs operating without authorisation under the BMW Rules in consultation with CBWTF. 17 occupiers have installed liquid waste treatment facility. ETP should be established in the remaining HCF for proper disposal of liquid waste.

10.11. Gujrat

No. of HCFs	35,804
No. of bedded HCFs	12,422
No. of non-bedded HCFs	23,572
No. of beds	2,23,997
No. of CBWTFs	20
No. of HCFs granted authorization	4,944
No. of HCFs utilizing CBWTFs	35,804
No. of HCFs having Captive Treatment Facilities	0
No. of Captive Incinerators Operated by HCFs	0
Quantity of BMW generated in kg/day	47,843
Quantity of BMW treated in kg/day	47,843
No. of HCFs/CBWTFs violated BMW Rules	1,024
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	1,024

Currently there is 20 CBWTF is present in the state and total generated BMW that is 47,843 kg/day is treated and disposed through CBWTF and no captive treatment facility present in the state. 2617 nos. occupiers are currently in operation without applying for the authorisation. 495 occupiers have installed liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report.

10.12. Haryana

No. of HCFs	7,107
No. of bedded HCFs	3,527
No. of non-bedded HCFs	3,580
No. of beds	67,332
No. of CBWTFs	11
No. of HCFs granted authorization	6,780
No. of HCFs utilizing CBWTFs	7,107
No. of HCFs having Captive Treatment Facilities	0
No. of Captive Incinerators Operated by HCFs	0
Quantity of BMW generated in kg/day	20,590
Quantity of BMW treated in kg/day	20,590
No. of HCFs/CBWTFs violated BMW Rules	169
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	169

Currently there is 11 CBWTF is present in the state and total generated BMW that is 20,590 kg/day is treated and disposed through CBWTF and no captive treatment facility present in the state. 169 nos. of occupiers are currently in operation without applying for the authorisation. 290 occupiers have installed liquid waste treatment facility and remaining 3237 HCFs are connected with terminal treatment facility and discharging into public sewer. SPCB should rectify the gaps identified in Annual Report.

10.13. Himachal Pradesh

No. of HCFs	9,365
No. of bedded HCFs	627
No. of non-bedded HCFs	8,738
No. of beds	18,824
No. of CBWTFs	3
No. of HCFs granted authorization	8,848
No. of HCFs utilizing CBWTFs	4,747
No. of HCFs having Captive Treatment Facilities	4,618
No. of Captive Incinerators Operated by HCFs	1
Quantity of BMW generated in kg/day	3769.5

Quantity of BMW treated in kg/day	3769.5
No. of HCFs/CBWTFs violated BMW Rules	44
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	44

Currently there are 03 CBWTF is present in the state and 3,612.5 kg/day of BMW is treated and disposed through CBWTF. 4618 of HCF in the state have captive treatment facility and 157.04 kg/day of BMW is treated and disposed through CTF. 315 occupiers are currently in operation without applying for authorisation. 8848 nos. of occupier have installed the liquid waste treatment facility. CPCB strongly recommend to address the identified gaps and non-compliances for proper BMWM.

10.14. Jharkhand

No. of HCFs	2,311
No. of bedded HCFs	1,368
No. of non-bedded HCFs	1,399
No. of beds	13,968
No. of CBWTFs	9
No. of HCFs granted authorization	160
No. of HCFs utilizing CBWTFs	1,434
No. of HCFs having Captive Treatment Facilities	4
No. of Captive Incinerators Operated by HCFs	4
Quantity of BMW generated in kg/day	9694.5
Quantity of BMW treated in kg/day	9011.2
No. of HCFs/CBWTFs violated BMW Rules	110
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	26

Currently there are 09 CBWTF is present in the state and 8843.8 kg/day of BMW is treated and disposed through CBWTF. 4 HCF in the state have captive treatment facility and 167.5 kg/day of BMW is treated and disposed through CTF. A gap of around 7% is observed between the generation and disposal of the BMW in the state. 140 occupiers are currently in operation without applying for authorisation. 177 nos. of occupier have installed the liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report.

10.15. Jammu and Kashmir

No. of HCFs	7,671
No. of bedded HCFs	2,924
No. of non-bedded HCFs	4,747
No. of beds	18,571

No. of CBWTFs	03
No. of HCFs granted authorization	556
No. of HCFs utilizing CBWTFs	2,430
No. of HCFs having Captive Treatment Facilities	0
No. of Captive Incinerators Operated by HCFs	0
Quantity of BMW generated in kg/day	8,651.42
Quantity of BMW treated in kg/day	8,418.53
No. of HCFs/CBWTFs violated BMW Rules	5,953
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	409

Currently there are only 3 CBWTF are operational in the state which seems less to cater the need of the state. A gap of around 2.7% between the generated and treated BMW is reported in the state. No captive treatment facility has been reported in the state. 6364 nos. of occupiers are currently in operation without applying for the authorisation. 104 occupiers have installed liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report.

10.16. Karnataka

No. of HCFs	48,431
No. of bedded HCFs	8,628
No. of non-bedded HCFs	37,765
No. of beds	2,31,007
No. of CBWTFs	25
No. of HCFs granted authorization	19,159
No. of HCFs utilizing CBWTFs	30,387
No. of HCFs having Captive Treatment Facilities	1,427
No. of Captive Incinerators Operated by HCFs	2
Quantity of BMW generated in kg/day	78,441
Quantity of BMW treated in kg/day	70,018
No. of HCFs/CBWTFs violated BMW Rules	2,075
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	667

Currently there are 25 CBWTF is present in the state and 6,66,934 kg/day of BMW is treated and disposed through CBWTF. 1,427 of HCF in the state have captive treatment facility and 3084 kg/day of BMW is treated and disposed through CTF. 3,733 occupiers are currently in operation without applying for authorisation. 10,808 nos. of occupier have installed the liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report.

.

10.17. Kerala

No. of HCFs	17,975
No. of bedded HCFs	1,694
No. of non-bedded HCFs	16,281
No. of beds	1,21,349
No. of CBWTFs	2
No. of HCFs granted authorization	16,550
No. of HCFs utilizing CBWTFs	17,489
No. of HCFs having Captive Treatment Facilities	44
No. of Captive Incinerators Operated by HCFs	14
Quantity of BMW generated in kg/day	62,122.78
Quantity of BMW treated in kg/day	62,122.78
No. of HCFs/CBWTFs violated BMW Rules	25
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	486

Currently there are 2 CBWTF is present in the state and 26,803 kg/day of BMW is treated and disposed through CBWTF. 44 of HCF in the state have captive treatment facility and 2,372 kg/day of BMW is treated and disposed through CTF. A gap of around 53% has been observed between the generated and treated BMW. 2148 occupiers are currently in operation without applying for authorisation. 10,811 nos. of occupier have installed the liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report.

10.18. Ladakh

No. of HCFs	400
No. of bedded HCFs	333
No. of non-bedded HCFs	67
No. of beds	1,042
No. of CBWTFs	0
No. of HCFs granted authorization	0
No. of HCFs utilizing CBWTFs	0
No. of HCFs having Captive Treatment Facilities	333
No. of Captive Incinerators Operated by HCFs	4
Quantity of BMW generated in kg/day	86.335
Quantity of BMW treated in kg/day	86.335
No. of HCFs/CBWTFs violated BMW Rules	0
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	0

Currently there is no CBWTF is operational in the UT. The generated BMW that is 86.3 kg/day is treated and disposed through captive treatment facility present with 333 HCF. There are 399

occupiers operating without applying for the authorisation. Only 1 occupier has installed liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report and expedite the process of establishment of CBWTF.

10.19. Lakshadweep

No. of HCFs	48
No. of bedded HCFs	09
No. of non-bedded HCFs	39
No. of beds	247
No. of CBWTFs	Nil
No. of HCFs granted authorization	48
No. of HCFs utilizing CBWTFs	0
No. of HCFs having Captive Treatment Facilities	10
No. of Captive Incinerators Operated by HCFs	0
Quantity of BMW generated in kg/day	86
Quantity of BMW treated in kg/day	86
No. of HCFs/CBWTFs violated BMW Rules	0
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	0

Currently there are no CBWTF present in the UT. The generated waste that is 86 kg/day is treated using captive treatment facility and deep burial pits by the HCF. No occupier is operating without applying for authorization. No occupier has installed the liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report and expedite the process of establishment of CBWTF.

10.20. Madhya Pradesh

No. of HCFs	10,371
No. of bedded HCFs	3,877
No. of non-bedded HCFs	6,494
No. of beds	1,35,695
No. of CBWTFs	14
No. of HCFs granted authorization	8493
No. of HCFs utilizing CBWTFs	7694
No. of HCFs having Captive Treatment Facilities	1
No. of Captive Incinerators Operated by HCFs	1
Quantity of BMW generated in kg/day	15631.8
Quantity of BMW treated in kg/day	15450.7
No. of HCFs/CBWTFs violated BMW Rules	872
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	782

Currently there are 14 CBWTF is present in the state and 15,387.7 kg/day of BMW is treated and disposed through CBWTF. Only 1 HCF in the state have captive treatment facility and 63.1 kg/day of BMW is treated and disposed through CTF. A gap of around 1.2% has been observed between the generated and treated BMW. 1630 occupiers are currently in operation without applying for authorisation. 3277 nos. of occupier have installed the liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report.

10.21. Maharashtra

No. of HCFs	70,083
No. of bedded HCFs	21,685
No. of non-bedded HCFs	48,398
No. of beds	3,39,351
No. of CBWTFs	29
No. of HCFs granted authorization	6,542
No. of HCFs utilizing CBWTFs	69,416
No. of HCFs having Captive Treatment Facilities	255
No. of Captive Incinerators Operated by HCFs	0
Quantity of BMW generated in kg/day	74,248
Quantity of BMW treated in kg/day	74,248
No. of HCFs/CBWTFs violated BMW Rules	643
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	607

Currently there are 29 CBWTF is present in the state and 74,084 kg/day of BMW is treated and disposed through CBWTF. 03 CBWTF are currently under construction in the state. 255 of HCF in the state have captive treatment facility and 164 kg/day of BMW is treated and disposed through CTF. 679 occupiers are currently in operation without applying for authorisation. 633 nos. of occupier have installed the liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report.

10.22. Manipur

No. of HCFs	754
No. of bedded HCFs	96
No. of non-bedded HCFs	658
No. of beds	4,854
No. of CBWTFs	1
No. of HCFs granted authorization	754
No. of HCFs utilizing CBWTFs	308
No. of HCFs having Captive Treatment Facilities	446
No. of Captive Incinerators Operated by HCFs	2
Quantity of BMW generated in kg/day	1442.9
Quantity of BMW treated in kg/day	1442.4
No. of HCFs/CBWTFs violated BMW Rules	1

No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	0

Currently there are 01 CBWTF is present in the state and 676 kg/day of BMW is treated and disposed through CBWTF. 446 of HCF in the state have captive treatment facility and 766.4 kg/day of BMW is treated and disposed through CTF. No occupier is reported to be currently in operation without applying for authorisation. 10 nos. of occupier have installed the liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report.

10.23. Meghalaya

No. of HCFs	1196
No. of bedded HCFs	183
No. of non-bedded HCFs	1,013
No. of beds	7,145
No. of CBWTFs	1
No. of HCFs granted authorization	1,056
No. of HCFs utilizing CBWTFs	24
No. of HCFs having Captive Treatment Facilities	283
No. of Captive Incinerators Operated by HCFs	1
Quantity of BMW generated in kg/day	2727.7
Quantity of BMW treated in kg/day	2427.7
No. of HCFs/CBWTFs violated BMW Rules	0
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	0

Currently there is 1 CBWTF is present in the state and 1902.31 kg/day of BMW is treated and disposed through CBWTF. 283 HCF in the state have captive treatment facility and 825.47 kg/day of BMW is treated and disposed through CTF. 140 occupiers are currently in operation without applying for authorisation. 38 nos. of occupier have installed the liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report.

10.24. Mizoram

No. of HCFs	407
No. of bedded HCFs	87
No. of non-bedded HCFs	320
No. of beds	3,280
No. of CBWTFs	0
No. of HCFs granted authorization	153
No. of HCFs utilizing CBWTFs	0
No. of HCFs having Captive Treatment Facilities	407
No. of Captive Incinerators Operated by HCFs	7

Quantity of BMW generated in kg/day	958
Quantity of BMW treated in kg/day	958
No. of HCFs/CBWTFs violated BMW Rules	0
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	0

Currently there are no CBWTF present in the state. 1 CBWTF is currently in the construction stage. The generated waste that is 958.06 kg/day is treated using captive treatment facility and deep burial pits by the HCF. No occupier is operating without applying for authorization. 88 occupiers have installed the liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report and expedite the process of establishment of CBWTF.

10.25. Nagaland

No. of HCFs	708
No. of bedded HCFs	251
No. of non-bedded HCFs	457
No. of beds	3,700
No. of CBWTFs	0
No. of HCFs granted authorization	636
No. of HCFs utilizing CBWTFs	0
No. of HCFs having Captive Treatment Facilities	5
No. of Captive Incinerators Operated by HCFs	5
Quantity of BMW generated in kg/day	1076.4
Quantity of BMW treated in kg/day	1076.4
No. of HCFs/CBWTFs violated BMW Rules	0
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	0

Currently there are no CBWTF present in the state. The generated waste that is 1076.4 kg/day is treated using captive treatment facility and deep burial pits by the HCF. No occupier is operating without applying for authorization. 12 occupiers have installed the liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report and expedite the process of establishment of CBWTF.

10.26. Odisha

No. of HCFs	4,007
No. of bedded HCFs	1,638
No. of non-bedded HCFs	2,369
No. of beds	4,854
No. of CBWTFs	6
No. of HCFs granted authorization	735
No. of HCFs utilizing CBWTFs	1275
No. of HCFs having Captive Treatment Facilities	2732

No. of Captive Incinerators Operated by HCFs	0
Quantity of BMW generated in kg/day	16677.6
Quantity of BMW treated in kg/day	16677.6
No. of HCFs/CBWTFs violated BMW Rules	0
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	0

Currently there is 06 CBWTF is present in the state and 5,945.1 kg/day of BMW is treated and disposed through CBWTF. 2732 HCF in the state have captive treatment facility and 10735.6 kg/day of BMW is treated and disposed through CTF. 34 occupiers are currently in operation without applying for authorisation. 88 nos. of occupier have installed the liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report.

10.27. Puducherry

No. of HCFs	390
No. of bedded HCFs	96
No. of non-bedded HCFs	294
No. of beds	52,140
No. of CBWTFs	1
No. of HCFs granted authorization	386
No. of HCFs utilizing CBWTFs	390
No. of HCFs having Captive Treatment Facilities	1
No. of Captive Incinerators Operated by HCFs	1
Quantity of BMW generated in kg/day	4706.4
Quantity of BMW treated in kg/day	4706.4
No. of HCFs/CBWTFs violated BMW Rules	8
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	3

Currently there is 01 CBWTF is present in the state and 4927 kg/day of BMW is treated and disposed through CBWTF. 01 HCF in the state have captive treatment facility and 15.4 kg/day of BMW is treated and disposed through CTF. No occupier is currently in operation without applying for authorisation. 31 nos. of occupier have installed the liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report.

10.28. Punjab

No. of HCFs	13,649
No. of bedded HCFs	3,998
No. of non-bedded HCFs	9,651
No. of beds	12,112
No. of CBWTFs	6

No. of HCFs granted authorization	2,464
No. of HCFs utilizing CBWTFs	11,139
No. of HCFs having Captive Treatment Facilities	0
No. of Captive Incinerators Operated by HCFs	0
Quantity of BMW generated in kg/day	19,841
Quantity of BMW treated in kg/day	19,841
No. of HCFs/CBWTFs violated BMW Rules	921
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	922

Currently there is 06 CBWTF is present in the state and total generated BMW that is 19,841 kg/day is treated and disposed through CBWTF and no captive treatment facility present in the state. 1003 nos. occupiers are currently in operation without applying for the authorisation. 10,841 HCFs have installed system for pre-treatment of liquid waste with 1-2% of sodium hypochlorite. However, 387 HCFs have provided ETP for the final treatment of liquid waste. SPCB should rectify the gaps identified in Annual Report.

10.29. Rajasthan

No. of HCFs	9,842
No. of bedded HCFs	6,744
No. of non-bedded HCFs	3,098
No. of beds	1,80,085
No. of CBWTFs	11
No. of HCFs granted authorization	2799
No. of HCFs utilizing CBWTFs	8546
No. of HCFs having Captive Treatment Facilities	505
No. of Captive Incinerators Operated by HCFs	0
Quantity of BMW generated in kg/day	19564.9
Quantity of BMW treated in kg/day	19564.9
No. of HCFs/CBWTFs violated BMW Rules	869
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	951

Currently there is 11 CBWTF is present in the state and 17986.975 kg/day of BMW is treated and disposed through CBWTF. 505 HCF in the state have captive treatment facility and 1577.93 kg/day of BMW is treated and disposed through deep burial pits. 677 occupiers are currently in operation without applying for authorisation. 315 nos. of occupier have installed the liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report.

10.30. Sikkim

No. of HCFs	319
No. of bedded HCFs	33
No. of non-bedded HCFs	286
No. of beds	2,329
No. of CBWTFs	0
No. of HCFs granted authorization	317
No. of HCFs utilizing CBWTFs	0
No. of HCFs having Captive Treatment Facilities	297
No. of Captive Incinerators Operated by HCFs	7
Quantity of BMW generated in kg/day	562
Quantity of BMW treated in kg/day	562
No. of HCFs/CBWTFs violated BMW Rules	0
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	0

Currently there are no CBWTF present in the state. The generated waste that is 562 kg/day is treated using captive treatment facility and deep burial pits by the HCF. 02 occupiers are operating without applying for authorization. 45 occupiers have installed the liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report and expedite the process of establishment of CBWTF.

10.31. Tamil Nadu

No. of HCFs	27,603
No. of bedded HCFs	7,922
No. of non-bedded HCFs	19,681
No. of beds	19,114
No. of CBWTFs	10
No. of HCFs granted authorization	27,353
No. of HCFs utilizing CBWTFs	27,603
No. of HCFs having Captive Treatment Facilities	0
No. of Captive Incinerators Operated by HCFs	0
Quantity of BMW generated in kg/day	49721.4
Quantity of BMW treated in kg/day	49721.4
No. of HCFs/CBWTFs violated BMW Rules	3
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	3

Currently there is 10 CBWTF is present in the state and total generated BMW that is 49,721 kg/day is treated and disposed through CBWTF and no captive treatment facility present in the state. 05 nos. occupiers are currently in operation without applying for the authorisation. 5904

occupiers have installed liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report.

10.32. Telangana

No. of HCFs	9,346
No. of bedded HCFs	4,830
No. of non-bedded HCFs	4,516
No. of beds	1,32,332
No. of CBWTFs	11
No. of HCFs granted authorization	5,036
No. of HCFs utilizing CBWTFs	9,346
No. of HCFs having Captive Treatment Facilities	0
No. of Captive Incinerators Operated by HCFs	0
Quantity of BMW generated in kg/day	25,306
Quantity of BMW treated in kg/day	25,306
No. of HCFs/CBWTFs violated BMW Rules	188
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	188

Currently there is 11 CBWTF is present in the state and total generated BMW that is 25,306 kg/day is treated and disposed through CBWTF and no captive treatment facility present in the state. 187 nos. occupiers are currently in operation without applying for the authorisation. 2093 occupiers have installed liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report

10.33. Tripura

No. of HCFs	1,960
No. of bedded HCFs	153
No. of non-bedded HCFs	1807
No. of beds	4768
No. of CBWTFs	01
No. of HCFs granted authorization	122
No. of HCFs utilizing CBWTFs	123
No. of HCFs having Captive Treatment Facilities	172
No. of Captive Incinerators Operated by HCFs	6
Quantity of BMW generated in kg/day	1975.87
Quantity of BMW treated in kg/day	1975.87
No. of HCFs/CBWTFs violated BMW Rules	0
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	0

Currently there is 01 CBWTF is present in the state and 182.8 kg/day of BMW is treated and disposed through CBWTF. 172 HCF in the state have captive treatment facility and 1793.1 kg/day of BMW is treated and disposed through captive treatment facility. No occupiers are currently in operation without applying for authorisation. 17 nos. of occupier have installed the liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report.

10.34. Uttarakhand

No. of HCFs	6,002
No. of bedded HCFs	1,705
No. of non-bedded HCFs	4,297
No. of beds	30,706
No. of CBWTFs	2
No. of HCFs granted authorization	4,707
No. of HCFs utilizing CBWTFs	2,711
No. of HCFs having Captive Treatment Facilities	2,690
No. of Captive Incinerators Operated by HCFs	2
Quantity of BMW generated in kg/day	7,113.3
Quantity of BMW treated in kg/day	7,101.6
No. of HCFs/CBWTFs violated BMW Rules	907
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	47

Currently there is 02 CBWTF is present in the state and 5814.9 kg/day of BMW is treated and disposed through CBWTF. 2690 HCF in the state have captive treatment facility and 1286.8 kg/day of BMW is treated and disposed through captive treatment facility. 1058 occupiers are currently in operation without applying for authorisation. 174 nos. of occupier have installed the liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report..

10.35. Uttar Pradesh

No. of HCFs	36145
No. of bedded HCFs	19301
No. of non-bedded HCFs	16844
No. of beds	3,56,321
No. of CBWTFs	23
No. of HCFs granted authorization	15,638
No. of HCFs utilizing CBWTFs	26,753
No. of HCFs having Captive Treatment Facilities	133
No. of Captive Incinerators Operated by HCFs	06
Quantity of BMW generated in kg/day	89210.65
Quantity of BMW treated in kg/day	89210.65

No. of HCFs/CBWTFs violated BMW Rules	149
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	149

Currently there are 23 CBWTF is present in the state and 89210.65 kg/day of BMW is treated and disposed through CBWTF. There are 03 CBWTF currently under construction. 133 HCF in the state have captive treatment facility and 358.25 kg/day of BMW is treated and disposed through captive treatment facility. 4729 occupiers are currently in operation without applying for authorisation. 202 nos. of occupier have installed the liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report.

10.36. West Bengal

No. of HCFs	9,712
No. of bedded HCFs	3,334
No. of non-bedded HCFs	6,378
No. of beds	1,45,750
No. of CBWTFs	7
No. of HCFs granted authorization	9,712
No. of HCFs utilizing CBWTFs	9,712
No. of HCFs having Captive Treatment Facilities	0
No. of Captive Incinerators Operated by HCFs	0
Quantity of BMW generated in kg/day	38,886.1
Quantity of BMW treated in kg/day	38,886.1
No. of HCFs/CBWTFs violated BMW Rules	149
No. of Show-Cause notices/Directions issued to defaulter HCFs/CBWTFs	149

Currently there are 07 CBWTF is present in the state and 38886.14 kg/day of BMW is treated and disposed through CBWTF. There are 8 CBWTF currently under construction. No HCF in the state have captive treatment facility. No occupiers are currently in operation without applying for authorisation. 963 nos. of occupier have installed the liquid waste treatment facility. SPCB should rectify the gaps identified in Annual Report.

S.No	Name of the State/UT	Total no. of Bedded Health Care Facilities (HCFs)	Total no. of Non-bedded Health Care Facilities (HCFs)	Total no. Health Care Facilities (HCFs)	Total no. of Beds	Authorization Status			No. of HCFs utilization CBWTFs	Total Quantity of BMW generated (kg/day)	Total Quantity of BMW Treated and Disposed (kg/day)	Captive BMW Treatment Facilities Operated by the (HCFs		Common Bio-medical Waste Treatment Facilities (CBWTFs)		Deep burial installed by HCF & CBWTFs		Total BMW treated by captive treatment facilities by HCF in Kg/day	Total BMW treated by CBWTFs (kg/day)	Total no. of violation by HCFs & CBWTFs	Total No. of show cause notices/Directions issued to defaulter HCFs/CBWTFs	No. of CBWTFs that have installed COEMS
						Total no. of HCFs applied for authorization	Total no. of HCFs granted authorization	Total no. of HCFs in operation without Authorization				No. of HCFs having Captive Treatment Facilities	No of Captive Incinerators Operated by HCFs	CBWTFs Operational	CBWTFs under Construction	HCF	CBWTFs					
I.	II.	III.	IV.	V.	VI.	VII.	VIII.	IX.	X.	XI.	XII.	XIII.	XIV	XV.	XVI.	XVII.	XVIII.	XIX.	XX	XXI	XXII.	XXIII.
1	Andaman Nicobar	53	219	272	1610	272	142	INP	0	734.79	734.79	56	6	0	0	50	0	734.79	0	130	130	nil
2	Andhra Pradesh	6779	6602	13381	162751	3077	3074	781	12208	16512.02	16512.02	0	nil	12	1	nil	nil	nil	16512.02	907	907	12
3	Arunachal Pradesh	189	137	326	2759	410	403	56	0	464.6	464.6	326	16	Nil	nil	205	Nil	464.6	Nil	56	56	nil
4	Assam	767	770	1537	30432	1011	516	526	315	8229.32	5208.93	597	28	2	nil	INP	Nil	3068.45	2140.48	482	81	1
5	Bihar	6061	20411	26472	104943	2571	2311	2811	7836	28246.3	13949.3	3	3	4	0	0	0	302.79	13646.51	2542	29	4
6	Chandigarh	44	899	943	4825	54	50	4	943	5710	5710	0	Nil	1	0	Nil	Nil	0	5710	0	0	1
7	Chhattisgarh	2292	4280	6705	50829	4642	4513	inp	3168	7504.9	7504.9	1735	2	5	3	ask	2	1225.78	6985.11	196	134	3
8	Daman &Diu and Dadra & Nagar Haveli	74	132	206	1449	108	79	23	206	324	324	0	0	Utilising CBWTF of Gujarat	0	0	0	0	324	2	2	Utilising Gujarat facility-OCEMS installed
9	Delhi	1141	9450	10591	60344	1548	1253	Nil	10591	29155	29155	1	nil	2	nil	nil	nil	402.21	29155	25	25	2
10	Goa	168	1035	1203	6573	405	241	109	1086	2128.4	2128.44	217	0	1	0	0	0	0	2128.46	336	0	1
11	Gujarat	12422	23572	35804	2,23,997	4883	4944	2617	35804	47843	47843	0	Nil	20	1	nil	nil	nil	47843	1024	1024	20
12	Haryana	3527	3580	7107	67332	6938	6780	169	7107	20590	20590	0	nil	11	nil	nil	nil	nil	20590	169	169	11
13	Himachal Pradesh	627	8738	9365	18824	9050	8848	315	4747	3769.51	3769.51	4618	1	3	1	4618	nil	157.04	3612.47	44	44	3
14	Jharkhand	1368	1399	2311	31968	654	160	140	1434	9694.5	9011.22	4	4	4	1			167.47	8843.76	110	26	3

S.No	Name of the State/UT	Total no. of Bedded Health Care Facilities (HCFs)	Total no. of Non-bedded Health Care Facilities (HCFs)	Total no. Health Care Facilities (HCFs)	Total no. of Beds	Authorization Status			No. of HCFs utilization CBWTFs	Total Quantity of BMW generated (kg/day)	Total Quantity of BMW Treated and Disposed (kg/day)	Captive BMW Treatment Facilities Operated by the (HCFs		Common Bio-medical Waste Treatment Facilities (CBWTFs)		Deep burial installed by HCF & CBWTFs		Total BMW treated by captive treatment facilities by HCF in Kg/day	Total BMW treated by CBWTFs (kg/day)	Total no. of violation by HCFs & CBWTFs	Total No. of show cause notices/Directions issued to defaulter HCFs/CBWTFs	No. of CBWTFs that have installed COEMS
						Total no. of HCFs applied for authorization	Total no. of HCFs granted authorization	Total no. of HCFs in operation without Authorization				No. of HCFs having Captive Treatment Facilities	No of Captive Incinerators Operated by HCFs	CBWTFs Operational	CBWTFs under Construction	HCF	CBWTFs					
I.	II.	III.	IV.	V.	VI.	VII.	VIII.	IX.	X.	XI.	XII.	XIII.	XIV	XV.	XVI.	XVII.	XVIII.	XIX.	XX	XXI	XXII.	XXIII.
15	J & K	2924	4747	7671	18571	1307	556	740	2430	8651.42	8418.53	0	Nil	3	1	INP	Nil	Nil	8418.53	6364	3	3
16	Karnataka	8628	37765	48431	231007	19265	19159	3733	30387	78441	70018	1427	2	25	2	Nil	18	3084	66934	2075	667	25
17	Kerala	1694	16281	17975	121349	16989	16550	2148	17489	62122.78	29175	44	14	2	nil	30	nil	2372.15	59750	25	486	2
18	Ladakh	333	67	400	1042	1	0	399	0	86.335	86.34	333	4	0	0	19	0	86.335	0	399	145	0
19	Lakshadweep	9	39	48	247	48	48	0	0	86	86	10	1	0	0	4	0	86	0	0	0	0
20	Madhya Pradesh	3877	6494	10371	135695	8741	8493	1630	7694	15631.86	15450.77	1	1	14	6	Nil	Nil	63.11	15387.66	872	782	14
21	Maharashtra	21685	48398	70083	339351	7181	6542	679	69416	74248	74248	255	0	29	3	1121	1	164	74084	M	607	28
22	Manipur	96	658	754	4854	754	754	nil	308	1442.92	1442.42	446	2	1	nil	INP	nil	766.42	676	1	nil	nil
23	Meghalaya	183	1013	1196	7077	1066	1066	130	42	2840.66	2840.66	400	1	1	0	375	0	926.59	1914.06	0	0	0
24	Mizoram	87	320	407	3280	153	153	0	0	958.06	958.06	407	7	0	0	401	0	958.06	0	0	0	0
25	Nagaland	251	457	708	3700	636	636	0	0	1076.43	1076.43	5	5	0	0	511	0	1076.43	0	0	0	0
26	Odisha	1638	2369	4007	52140	1088	735	34	1275	16677.6	16677.64	2732	0	6	0	2732	2	10732.56	5945.08	100	100	6
27	Puducherry	96	294	390	12112	390	386	0	390	4706.4	4706.4	1	1	1	0	0	0	15.4	4927	8	3	1
28	Punjab	3998	9651	13649	79954	3092	2464	1003	11139	19841	19841	0	nil	6	nil	nil	nil	nil	19841	921	922	6
29	Rajasthan	6744	3098	9842	180085	2801	2799	677	8546	19564.9	19564.91	505	0	11	1	505	0	1577.93	17986.975	869	951	11
30	Sikkim	33	286	319	2329	317	317	2	0	562	562	297	7	0	0	179	0	562	0	0	0	NA
31	Tamil Nadu	7922	19681	27603	19114	27598	27353	5	27603	49721.43	49721.43	0	Nil	10	1	Nil	Nil	0	49721.43	3	3	10
32	Telengana	4830	4516	9346	132332	5238	5036	187	9346	25306	25306	0	NIL	11	0	NIL	Nil	NIL	25306	188	188	11

S.No	Name of the State/UT	Total no. of Bedded Health Care Facilities (HCFs)	Total no. of Non-bedded Health Care Facilities (HCFs)	Total no. Health Care Facilities (HCFs)	Total no. of Beds	Authorization Status			No. of HCFs utilization CBWTFs	Total Quantity of BMW generated (kg/day)	Total Quantity of BMW Treated and Disposed (kg/day)	Captive BMW Treatment Facilities Operated by the (HCFs		Common Bio-medical Waste Treatment Facilities (CBWTFs)		Deep burial installed by HCF & CBWTFs		Total BMW treated by captive treatment facilities by HCF in Kg/day	Total BMW treated by CBWTFs (kg/day)	Total no. of violation by HCFs & CBWTFs	Total No. of show cause notices/Directions issued to defaulter HCFs/CBWTFs	No. of CBWTFs that have installed COEMS
						Total no. of HCFs applied for authorization	Total no. of HCFs granted authorization	Total no. of HCFs in operation without Authorization				No. of HCFs having Captive Treatment Facilities	No of Captive Incinerators Operated by HCFs	CBWTFs Operational	CBWTFs under Construction	HCF	CBWTFs					
I.	II.	III.	IV.	V.	VI.	VII.	VIII.	IX.	X.	XI.	XII.	XIII.	XIV	XV.	XVI.	XVII.	XVIII.	XIX.	XX	XXI	XXII.	XXIII.
33	Tripura	153	1807	1960	4768	122	122	0	123	1975.87	1975.87	172	6	1	0	166	0	1793.1	182.77	0	0	1
34	Uttarakhand	1705	4297	6002	30706	4914	4707	1058	2711	7113.37	7101.60	2690	2	2	2	ask	0	1286.8	5814.8	907	47	2
35	Uttar Pradesh	19301	16844	36145	356321	17340	15638	4729	26753	89210.65	89210.65	133	10	23	3	132	0	358.25	88851.75	3402	611	20
36	West Bengal	3334	6378	9712	145750	9712	9712	0	9712	38886.14	38886.14	0	0	7	8	0	0	0	38886.14	149	149	7
37	DGAFMS	226	471	697	38640	97	97	0	0	5152.3	5152.3	75	30	0	0	45	0	926.586	4225.72	0	0	0
	total	**125259**	**267155**	**393939**	**2465063**	**164473**	**156637**	**24705**	**310809**	**705209.569**	**645411.851**	**17490**	**153**	**218**	**34**	**11093**	**23**	**33358.829**	**646343.75**	**22306**	**8291**	**208**